

Quantifying the Sensitivity of Quantitative Trade Models*

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Abstract

The core ingredient of modern quantitative trade models is cross-country heterogeneity in productive capabilities across goods. In workhorse models this heterogeneity is assumed to follow specific functional forms—such as i.i.d. Frechet or Pareto—but little is known about how these models’ counterfactual predictions would change under alternative distributional assumptions. We fill this gap by estimating sharp bounds on counterfactuals across all distributions within given neighborhoods of the workhorse model. Importantly, our method only considers distributions that can generate baseline equilibria that exactly match all trade flow data points, and hence reproduce empirical facts such as the amount of openness, the fit of the gravity equation, and the estimated trade elasticity. Our estimated bounds on the gains from trade are wide, even for small departures from the workhorse model: the upper bound is more than four times that of the workhorse model (so trade can be much more valuable than typically thought) and the lower bound is zero (so trade can even generate no gains). Imposing high-level restrictions on the distributions considered—for example, maintaining Frechet/Pareto marginals but relaxing independence, or vice versa—does little to shrink these bounds.

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1 Introduction

A major advance in the field of international economics has come from the development of quantitative trade models: general equilibrium models that are designed to match empirical patterns about factual scenarios and make predictions about economic responses to counterfactual ones.¹ A core ingredient of these models is the extent to which productive capabilities in the model are heterogeneous across goods and countries. Indeed, the exploitation of this heterogeneity—where countries import the goods that cannot be made cheaply at home—is what gives rise to trade in the first place. However, quantifying this heterogeneity poses a foundational challenge: how can one hope to measure a country’s ability to produce the goods it imports and doesn’t produce?

To make progress, researchers have posited specific functional forms to describe production heterogeneity around the world. Two choices that result in so-called “gravity models”—the Frechet distribution used by [Eaton and Kortum \(2002\)](#) for competitive models and the Pareto distribution used by [Chaney \(2008\)](#) for monopolistically competitive models—have become enormously influential for workhorse quantitative modeling. But how would economists’ predictions about counterfactuals, such as the size of the gains from trade, change if alternative forms of production heterogeneity were considered instead?

In this paper we conduct a comprehensive quantification of the sensitivity of the gains from trade in quantitative trade models as we allow for departures from these workhorse assumptions. Our approach embodies two key constraints. First, in order to focus on the role of production heterogeneity, we hold constant all other model components, such as preferences, factor market equilibrium, and market conduct. Second, we ensure that the standard method for gravity model estimation—the same orthogonality condition, the same ability to match all bilateral trade flow values perfectly, the same empirical fit of the estimated gravity equation, and the same elasticity of trade with respect to tariffs—is applied to every model variant that we consider. Then, subject to these constraints, we calculate maximal and minimal bounds across sets of underlying productivity distributions that start “close” to the workhorse versions and then grow increasingly large. Our main finding is that the gains from trade in quantitative models are highly sensitive to alternative distributions of production heterogeneity.

To interpret this result, begin with the case of complete agnosticism about the distribu-

¹Two recent chapters in the *Handbook of International Economics*, [Costinot and Rodríguez-Clare \(2014\)](#) and [Caliendo and Parro \(2022\)](#), provide surveys of this practice. Analogous progress on quantitative spatial models of intra-national trade is also surveyed in two recent chapters of the *Handbook of Regional and Urban Economics*, [Allen and Arkolakis \(2025\)](#) and [Redding \(2025\)](#).

tion of production heterogeneity across countries and goods. Our new analytical bounds on the gains from trade for this case, which are identical across the competitive and monopolistically competitive models, have a lower bound of zero and an upper bound that is given by a formula involving one data ingredient (the openness of the country in question) and one parameter (the elasticity of substitution across goods in consumer preferences).² Considering the case of France, a country with representative levels of openness among those in our sample, we find that the all-distributions upper bound on that country’s gains from trade is 4.4 times larger than the typical gravity model benchmark. Complete agnosticism about production heterogeneity therefore leads to wide bounds on predictions about this important object.

While these wide bounds allow for distributions that may be arbitrarily different from those of the gravity model workhorse (the Frechet distribution in [Eaton and Kortum \(2002\)](#) and the Pareto distribution in [Chaney \(2008\)](#)), our next result suggests that similarly wide bounds are possible even for distributions that are restricted to be “close” to these benchmarks. We employ a standard divergence measure of the similarity of any two probability distributions and emphasize a level of this divergence that is similar to the one implicitly used when analysts perform standard robustness checks on workhorse models. Even distributions within this restricted set imply that France’s gains from trade can range between zero and 3.9 times the gravity model value, so they can essentially span most of the range (i.e. zero to 4.4 times) that exists when all distributions are allowed for.

These findings suggest that the gains from trade can be much larger than typically thought, even when confined to data-consistent models with productivity heterogeneity that is distributed similarly to that in its gravity model formulation. On the other hand, these confines also contain models with almost no gains at all.

Evaluating the infinitely large set of candidate distributions within any given neighborhood is of course impossible. And matters are harder still when any candidate distribution must be tested for whether it can generate a global equilibrium whose trade flows match all data points and that it is consistent with standard gravity model estimation requirements. To circumvent this challenge we draw on advances in the field of distribution-robust econometrics. In particular, [Hansen and Sargent \(2001\)](#) and [Christensen and Connault \(2023\)](#) consider an optimization problem over all distributions of a

²[Arkolakis et al. \(2012\)](#) derive a similar formula—involving the target country’s openness and a model parameter—to quantify the gains from trade in the gravity model. But in their case the model parameter is the trade elasticity, which corresponds to a parameter of the particular productivity distribution in the gravity model in the [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) cases. Our formula instead describes the upper bound on the gains from trade over all such distributions and the parameter that matters derives from preferences alone.

random variable within a given divergence neighborhood, where the objective and constraints are expectations of (known functions of) the random variable. They show that this infinite-dimensional program has a dual that is finite-dimensional, with dimension given by the number of constraints in the original problem.

Following this logic, we write our problem—maximize gains from trade subject to the constraints of model equilibrium and matching all trade data—as one with an inner loop (given all unknown model parameters and equilibrium variables in the factual and counterfactual equilibria) and an outer loop (that searches over these parameters and equilibrium variables). The inner loop is infinite-dimensional but benefits from the finite-dimensional dual formulation in [Christensen and Connault \(2023\)](#). And the outer loop is too high-dimensional to be feasible but benefits from dimensionality-reduction techniques that we develop for the gains-from-trade problem. Put together, the program becomes feasible.

Given that we find so little robustness of quantitative models, even for small amounts of departure from the gravity model, it is natural to seek to restrict the admissible productivity distributions further. Our final set of results does this by imposing high-level restrictions that are guided by three core features of the benchmark distribution used in gravity models: (i) countries take draws independently from one another; (ii) countries all have the same marginal distribution; and (iii) the particular marginal distribution that is common is the Frechet form in [Eaton and Kortum \(2002\)](#) and the Pareto form in [Chaney \(2008\)](#). We show how to extend our baseline analysis—varying the model’s production heterogeneity while matching all trade flows and standard orthogonality conditions—to incorporate constraints (i) through (iii), one at a time or in combination, in order to learn which particular features of the benchmark distribution do the most to constrain the model’s predictions. For example, constraint (i) can be approximated by adding thousands of constraints in order to mandate that, for every pair of countries, and at any chosen quantile, the expected value of the product of the countries’ productivity draws is equal to the product of the countries’ own expectations.

Across a wide range of restrictions imposed, the broad message that emerges is that gains from trade bounds still remain wide. We begin by establishing that for the case of monopolistically competitive models these restrictions have no direct bite at all. Turning to the competitive case, our distribution-agnostic upper bound formula continues to hold despite adding requirements such as independence or common (and Frechet) marginals on the set of distributions that are permitted. We then move on to evaluate neighborhoods that are “close” to the gravity model. In this setting the largest possible gains from trade do shrink in the face of high-level restrictions, but the effects are modest. For exam-

ple, when we impose conditions that approximate full independence our upper bound (at fixed neighborhood size) shrinks from 3.9 times the baseline to 3.3 times the baseline (i.e. by 18%). And this is similar for the case where we impose zero correlation (18% shrinkage), common but flexible marginals (11%), common and [Eaton and Kortum \(2002\)](#) marginals (11%), and the joint imposition of independence and common but flexible marginals (24%).

To summarize, even in the most restricted version that we allow for—that with perfect competition and where countries draw productivities independently and identically from some common distribution—there exist versions of the productivity distribution that imply gains from trade that can be either zero or 3.2 times the value in workhorse gravity models, despite being an otherwise conventional trade model that matches all available data and standard orthogonality conditions.

We believe these results are surprising. One might conjecture that once key facts about trade flows are matched—facts such as the extent of openness, the impressive fit of the gravity equation, and the elasticity of trade with respect to tariffs—there may be little room to maneuver in search of alternative models with distributions of production heterogeneity and yet considerably different gains from trade. To the contrary, our findings demonstrate that the empirical fit of the gravity equation has little connection to the extent to which any given country’s trade flows will respond to tariff restrictions, and hence about the shape of that country’s import demand curve (the area under which quantifies the welfare losses from an inability to trade). This remains true even when all countries have the same import demand curve, as they do in the case where productivities are drawn independently and identically around the world.

Related literature. Our analysis builds on previous explorations of how quantitative trade models behave under varying assumptions about production heterogeneity. In particular, our approach is heavily influenced by that of [Arkolakis et al. \(2012\)](#), who proposed that the sensitivity of conclusions obtained across models should be conducted while conditioning on two requirements: (i) that the models are calibrated to match the same underlying trade data; and (ii) that the models agree on the value of the trade elasticity because they are estimated by assuming the same underlying orthogonality condition. Indeed, those authors show that, once these two conditions hold, counterfactuals are strikingly robust across the four types of production heterogeneity they study—the Frechet distribution of [Eaton and Kortum \(2002\)](#) and the degenerate distribution of [Armington \(1969\)](#) in the competitive case, and the Pareto distribution of [Chaney \(2008\)](#) and the degenerate distribution of [Krugman \(1980\)](#) in the monopolistically competitive case. We maintain

these same two requirements but then broaden the set of productivity distributions considered to include all such distributions in order to fully assess the robustness uncovered by [Arkolakis et al. \(2012\)](#).

Our assessment is therefore complementary to other work that has evaluated productivity distributions that depart from the four members of the [Arkolakis et al. \(2012\)](#) class. On the competitive side, for example, [Lind and Ramondo \(2023\)](#) emphasize productivity distributions in which countries' draws exhibit rich forms of dependence. We embrace this direction by considering distributions that feature arbitrary forms of dependency but are constrained to have the same set of Frechet marginals as in the workhorse model. [Adao et al. \(2017\)](#) demonstrate how the implications of underlying productivity heterogeneity are always summarized by "reduced" import demand systems, and that those demand systems can be estimated (from a cross-section) directly as long as the underlying system is assumed to be identical across countries.³ Models with productivity distributions that feature cross-country independence and common marginals deliver import demand systems that are indeed identical across countries, so we pursue the direction proposed by [Adao et al. \(2017\)](#) by evaluating bounds under this restriction.

Turning to monopolistic competition, several studies have introduced models where countries' productivities continue to be drawn independently, as in the gravity model, but from marginal distributions that depart from the case of common Pareto. For example, [Helpman et al. \(2004\)](#) examine the case of country-specific Pareto distributions; [Helpman et al. \(2008\)](#), [Melitz and Redding \(2015\)](#), and [Feenstra \(2018\)](#) consider the truncated Pareto case; [Head et al. \(2014\)](#), [Bas et al. \(2017\)](#), and [Fernandes et al. \(2023\)](#) analyze the case of log-normal distributions; [Nigai \(2017\)](#) proposes a mixture of log-normal and Pareto; and [Erhardt et al. \(2021\)](#) and [Adao et al. \(2026\)](#) allow for nonparametric distributions estimated separately by country.⁴ These papers often use micro-data to estimate features of the firm- or good-level productivity distribution in question. But, as we show below, inferences about any country's gains from trade always involve extrapolating from the distribution of goods that are produced at baseline (on which data is available) to goods that are not (and hence for which data is always missing). Our analysis therefore bounds what is possible from such extrapolation.

³An advantage of the reduced demand system approach in [Adao et al. \(2017\)](#) is that it summarizes the joint role played by technology and tastes for generating trade. Our analysis holds preferences constant across all models considered so it only relates to the technology side of their analysis.

⁴Several of these studies also follow [Eaton et al. \(2011\)](#) in allowing for additional forms of production-side heterogeneity beyond that in [Melitz \(2003\)](#), such as heterogeneity in fixed costs of exporting in [Fernandes et al. \(2023\)](#) and [Adao et al. \(2026\)](#). Our analysis sticks to the [Melitz \(2003\)](#) case.

2 Quantitative trade model environments

In this section we describe the core economic environment of a multi-country, single-factor trade model with many goods and arbitrary distributions of productivities (across goods and countries) around the world. We then describe how one can conduct counterfactual exercises (such as computing the gains from trade for any country) in such a model. Finally, we describe the version of this model that obtains when the global productivity distribution takes a particular form, as in workhorse versions of the model. Our presentation of these elements begins with a competitive version of this model (i.e. a Ricardian model of international trade) and then a monopolistically competitive version (i.e. a [Melitz \(2003\)](#) model of international trade).

2.1 A competitive model of trade

Setup. There are D countries, potentially trading any of the goods along a continuum indexed by $\omega \in [0, 1]$. Each country d produces goods using a factor of production that is owned by a representative household there and is immobile across countries.

Demand. The representative household in each country d has preferences that take the CES form across the continuum of goods:

$$U_d = \left(\int_0^1 q_d(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}},$$

with σ capturing the elasticity of substitution and $q_d(\omega)$ denoting the quantity of good ω consumed in country d . As is standard, we work with the case in which goods are substitutes (i.e. $\sigma > 1$).⁵

We let $p_{od}(\omega)$ denote the price of good ω in country d , if it were to be shipped from country o to country d . The determination of these prices involves the supply side of the economy, introduced below. But regardless of such determination, utility-maximization implies that the expenditure in d on good ω from country o will be

$$X_{od}(\omega) = E_d \gamma_d^{-1} p_{od}(\omega)^{1-\sigma}, \quad (1)$$

⁵This restriction is needed to ensure that the gains from trade, discussed below, always remain finite as we consider alternative technologies in which d may be unable to produce some goods.

where E_d is total expenditure in d and where

$$\gamma_d \equiv \sum_o \int_0^1 p_{od}(\omega)^{1-\sigma} d\omega, \quad (2)$$

is a transformation of the standard CES price index. In particular, the real income of the representative household in d is given by $W_d = E_d \gamma_d^{\frac{1}{\sigma-1}}$.

Supply. The core of the Ricardian model lies on the supply side. We assume that producers in country o are endowed with a technology with productivity $z_o(\omega)$ for producing any good ω . The marginal cost of a producer of ω in o when selling in country d is given by

$$MC_{od}(\omega) \equiv \frac{w_o(1 + t_{od})}{A_{od}z_o(\omega)}, \quad (3)$$

where w_o is the cost of a unit of the factor of production in o , t_{od} denotes the ad-valorem tariff charged in d on imports from o , and A_{od} is a bilateral productivity shifter.⁶ The productivity level $z_o(\omega)$ augments production by shifting downwards the marginal cost of producing ω in o .

Because the good ω is homogeneous, perfect competition implies that prices in location d are determined by the marginal cost of the lowest-cost supplier to that location. In particular, the price of the version of good ω that hails from o and is sold in d must satisfy

$$p_{od}(\omega) = \begin{cases} MC_{od}(\omega) & \text{if } o = \arg \min_{o' \in D} MC_{o'd}(\omega), \\ \infty & \text{otherwise.} \end{cases} \quad (4)$$

Naturally, consumers in country d only buy from the minimum-price supplier of any good; that is, $p_{od}(\omega) = \infty$ implies $X_{od}(\omega) = 0$ in equation (1).

Equilibrium. We let L_o denote the (exogenous) supply of the factor of production in o , meaning that total factor income in o is given by $w_o L_o$. We assume that tariff revenue is spent on projects that do not directly contribute to utility.⁷ To account for trade imbalances, we define country d 's trade deficit as the difference between expenditure and income $B_d \equiv E_d - w_d L_d$. A competitive global equilibrium in this model is then one in

⁶The shifter A_{od} incorporates (inversely) an “iceberg” transport cost that acts like a tariff but generates no tax revenue.

⁷This assumption simplifies the formulas presented below, which makes our core focus—the role of productivity heterogeneity—clearer. In practice, the share of tariff revenue in expenditure is negligible for the countries in our application.

which the factor market clears,

$$w_o L_o = \sum_d \int_0^1 \frac{1}{1 + t_{od}} X_{od}(\omega) d\omega \quad (5)$$

in every country o .

2.2 Stochastic representation

The key ingredient of this competitive model is the joint distribution of countries' productivities, $z_o(\omega)$ for all $o \in D$. For example, if productivities were to be distributed in such a way that they are perfectly correlated across countries then no country would have a comparative advantage in any good and there would be no trade (or gains from trade). On the other hand, if countries' distributions were relatively uncorrelated, or even negatively correlated, then a rationale for international specialization and hence trade emerges in the competitive equilibrium.

To emphasize the role of such distributions we follow the approach in [Eaton and Kortum \(2002\)](#) and treat the productivity $z_o(\omega)$ as a random variable that is drawn from a multi-country stochastic process. However, in a departure from such work we allow for an *arbitrary* joint cumulative distribution function F . That is, F denotes a function $F(\{z\})$ that equals the probability that the D -dimensional vector of random variables $\{z_o(\omega)\}$ lies below the D -dimensional vector $\{z\}$. Because F is the joint distribution across countries this notation allows for two important possibilities: that countries have different marginal distributions (e.g. that country o tends to get more dispersed draws than country o' does) and that countries' draws display forms of dependence (e.g. that the draws of country o are negatively correlated with those of country o').

Using this stochastic representation we let the random variable $p_{od}(\omega)$ be defined as a function of $z_o(\omega)$ using equations (3) and (4). Similarly, the random variable $X_{od}(\omega)$ is defined, via equation (1), as a function of $p_{od}(\omega)$. Finally, we follow the law-of-large-numbers convention of using the expectation operator $\mathbb{E}_F[\cdot]$ over the random variable $\{z_o(\omega)\}$ instead of integrals across the continuum of goods. Doing so, we can re-write the endogenous variable γ_d from equation (2) as

$$\gamma_d = \sum_o \mathbb{E}_F \left[p_{od}(\omega)^{1-\sigma} \right], \quad (6)$$

and the market clearing condition (5) as

$$w_o L_o = \sum_d \frac{1}{1 + t_{od}} \mathbb{E}_F [X_{od}(\omega)]. \quad (7)$$

For future reference, we denote the total value of bilateral trade as $X_{od} \equiv \mathbb{E}_F [X_{od}(\omega)]$ and the total bilateral import share in expenditure as $\lambda_{od} \equiv X_{od} / E_d$.

2.3 Counterfactuals

Our goal is to study how economic conditions in any given country d respond to counterfactual changes in the exogenous primitives of this model. In particular, we consider a change from the initial set of tariffs $\{t\}$ to the counterfactual set $\{t'\}$, and do so while holding constant all other exogenous model features—endowments $\{L\}$, bilateral productivity shifters $\{A\}$, preferences σ , the deficit to GDP ratio $\{B/Y\}$, and the multi-country productivity distribution F .

The counterfactual equilibrium is one in which all endogenous variables have changed correspondingly. This means that each producer's counterfactual marginal cost is $MC'_{od}(\omega) = w'_o(1 + t'_{od})A_{od}^{-1}[z_o(\omega)]^{-1}$ and the counterfactual price $p'_{od}(\omega)$ follows from (4) but evaluated at $MC'_{od}(\omega)$. Similarly, counterfactual expenditures are

$$X'_{od}(\omega) = (w'_d L_d + B'_d)(\gamma'_d)^{-1} p'_{od}(\omega)^{1-\sigma}, \quad (8)$$

$$\text{with } \gamma'_d = \sum_o \mathbb{E}_F [p'_{od}(\omega)^{1-\sigma}], \quad (9)$$

and equilibrium factor prices clear markets in the counterfactual equilibrium

$$w'_o L_o = \sum_d \frac{1}{1 + t'_{od}} \mathbb{E}_F [X'_{od}(\omega)]. \quad (10)$$

Gains from trade. A key counterfactual exercise of interest concerns the gains from trade for country d . This is calculated by changing tariffs from their factual values of $\{t\}$ to the counterfactual tariffs $\{t'\}$ that would place d in autarky (i.e. $t_{od} = \infty$ for all $o \neq d$) and then solving for the proportional change in real income. That is, we define

$$GT_d \equiv 1 - \left(\frac{w'_d L_d + B'_d}{w_d L_d + B_d} \right) \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}}. \quad (11)$$

Given the numeraire choice of $w_d = w'_d = 1$, and using the convention that d 's deficit is fixed as a share of its nominal GDP, this further simplifies to

$$GT_d = 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}}. \quad (12)$$

Additional sets of counterfactual calculations are also feasible.⁸

2.4 The **Eaton and Kortum (2002)** competitive model

Our presentation of the competitive Ricardian model so far has held for any given distribution of global productivities F . However, one particularly tractable version of F , proposed in the seminal model of **Eaton and Kortum (2002)**, is that where countries' productivity levels are drawn independently from a Frechet distribution with scale parameter equal to one and shape parameter equal to $\theta^* > 0$, i.e. $z_o(\omega) \sim \text{Frechet}(1, \theta^*)$ for all o .⁹ We refer to this distribution as F^* in what follows (for the subset of our analysis referring to the competitive model) and treat it as a key reference point with which to compare the competitive model based on other distributions F .

It is worth emphasizing that F^* imposes three separate types of restrictions relative to the unrestricted F : (i) countries' distributions are independent from one another, so the joint global distribution is the product of country-specific marginal distributions; (ii) those country-specific marginal distributions are all common; and (iii) the common marginal distribution takes the particular form of $\text{Frechet}(1, \theta^*)$.

2.5 Monopolistic competition and the **Melitz (2003)** model

Having described the general Ricardian model of trade under perfect competition, along with its **Eaton and Kortum (2002)** special case, we turn now to the case of monopolistic competition a la **Melitz (2003)**. This is similar in many respects to the Ricardian model, so our discussion will be brief.

⁸In particular, for any known function $k(\cdot)$, the methods described below can be applied to any counterfactual calculation that takes the form $\kappa = \mathbb{E}_F[k(\{z(\omega)\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}, \{t\}, \{t'\}, \{L\}, \{L'\}, \{A\}, \sigma)]$. This includes the effects of expanded sets of shocks (such as trade liberalization, technological change, and endowment growth) on expanded sets outcomes (such as trade flows, wages, terms of trade, etc.).

⁹That is, $F^*(\{z\}) = \prod_o F_o^*(z)$, where $F_o^*(z) = \Pr(z_o(\omega) \leq z) = e^{-z^{-\theta^*}}$. We note that some authors, including **Eaton and Kortum (2002)**, define $F_o^*(z)$ to include a scale parameter $T_o > 0$ that is o -specific, with $F_o^*(z) = e^{-T_o z^{-\theta^*}}$, rather than $T_o = 1$ as in our convention. Because we include the free parameters A_{od} in (3), and these parameters enter the model symmetrically to $z_o(\omega)$, our choice of $T_o = 1$ is without loss.

Model setup. This model continues to feature D countries in which the representative household has CES preferences and hence demand follows equation (1). Production continues to be supplied by producers with marginal costs $MC_{od}(\omega)$ given by equation (3), on the basis of (variable) productivity draws $z_o(\omega)$ from an *arbitrary* global distribution that we continue to denote by F .

However, there are two differences relative to the Ricardian model. First, market conduct is now monopolistically (rather than perfectly) competitive, and there are fixed costs of producing. Monopolistically competitive conduct implies (in this CES demand case) that if a producer from o sells good ω in d then the price $p_{od}(\omega)$ will be marked up over the producer's marginal cost by the amount $\sigma/(\sigma - 1)$. In addition, a bilateral fixed cost $w_o f_{od}$ is incurred whenever a producer from o sells in d , so production will only take place where variable profits exceed such fixed costs.¹⁰ Collecting these observations we have

$$p_{od}(\omega) = \begin{cases} \frac{\sigma}{\sigma-1} MC_{od}(\omega) & \text{if } X_{od}(\omega) > \sigma f_{od} w_o, \\ \infty & \text{otherwise.} \end{cases} \quad (13)$$

The inequality $X_{od}(\omega) > \sigma f_{od} w_o$ in (13) is equivalent to a condition that productivity exceeds an endogenous pair-specific cutoff, i.e., $z_o(\omega) > \hat{z}_{od}^*$. We impose two standard restrictions on the parameters that give rise to the baseline equilibrium. First, we assume that parameters are such that $\hat{z}_{od}^* \geq \hat{z}_{oo}^*$ for all $d \neq o$, which ensures that selection into exporting is always active in the baseline. Second, we assume that all cutoffs are interior—i.e., for all od , there exists some mass of goods for which $z_o(\omega) > \hat{z}_{od}^*$ and some mass of goods for which $z_o(\omega) < \hat{z}_{od}^*$ —which ensures that some firms are active in each od pair in the baseline.¹¹

The other main difference brought about by monopolistic competition is that the equilibrium mass N_o of potential producers in o who choose to take a productivity draw $z_o(\omega)$ from F is endogenous. We assume that taking such a draw costs $w_o f_{Eo}$ and that a free entry condition mandates that the global profits a producer from o will obtain, in expectation, after drawing its productivity will be equal to the cost $w_o f_{Eo}$. This implies:

$$\sum_d \mathbb{E}_F \left[\max\{0, \frac{1}{\sigma} X_{od}(\omega) - w_o f_{od}\} \right] = w_o f_{Eo}. \quad (14)$$

Since aggregate profits are zero, household income in o continues to be given by $Y_o =$

¹⁰As is standard, we assume that, in equilibrium, firms produce differentiated products. This means that there is only one producer per good ω in any destination market d .

¹¹A sufficient condition for this is that the upper bound of the support of F is ∞ .

$w_o L_o$.

The market clearing and price index expressions of (6) and (7) are similar to the competitive model, except that they now include N_o ,

$$\gamma_d = \sum_o N_o \mathbb{E}_F \left[p_{od}(\omega)^{1-\sigma} \right], \quad (15)$$

$$w_o L_o = \sum_d \left(\frac{\sigma-1}{\sigma} \frac{1}{1+t_{od}} + \frac{1}{\sigma} \right) N_o \mathbb{E}_F [X_{od}(\omega)]. \quad (16)$$

Finally, we note that, in this model, total bilateral trade is given by $X_{od} = N_o \mathbb{E}_F [X_{od}(\omega)]$.

Summarizing, equilibrium in the monopolistically competitive model corresponds to a vector of global wages $\{w\}$, price indices $\{\gamma\}$, and masses of potential producers $\{N\}$ such that equations (1), (3) and (13)-(16) hold for all countries.

Gains from trade counterfactuals. Simulations of scenarios involving counterfactual tariffs $\{t'\}$, such as those that implement autarky for a given country of interest d , can be computed using a similar procedure to that discussed for the Ricardian model above. That is, we hold constant all exogenous primitives—endowments $\{L\}$, bilateral productivity shifters $\{A\}$, bilateral fixed costs of production $\{f\}$, entry costs $\{f_E\}$, preferences σ , and the global distribution of productivities F —and solve for the new equilibrium at counterfactual $\{t'\}$ in which all endogenous variables—wages $\{w\}$, price indices $\{\gamma\}$ and masses of potential firms $\{N\}$ —adjust to the new equilibrium. We then compute the gains from trade for any country d following again the formula in equation (12), where $\{t'\}$ implements autarky for country d .

Reference distribution from Chaney (2008). As with the Ricardian model, our presentation of the monopolistic competition model à la Melitz (2003) has allowed for the global distribution of productivity levels F to take any form. However, since the pioneering work of Chaney (2008), it has become common to deploy one particularly tractable distribution: that in which each country o draws its productivity $z_o(\omega)$ independently from a Pareto distribution with scale parameter equal to one and a shape parameter equal to $\theta^* > 0$.¹² We therefore take this to be the reference distribution, denoted F^* , when studying the monopolistic competition model. As with the Ricardian model, relative to the unrestricted distribution F , the reference distribution F^* imposes three separate types of restrictions: full independence, common marginals, and particular (i.e. $\text{Pareto}(1, \theta^*)$) marginals.

¹²That is, $F^*(\{z\}) = \prod_o F_o^*(z)$, where $F_o^*(z) = \Pr(z_o(\omega) \leq z) = 1 - z^{-\theta^*}$.

3 Estimating the gains from trade

Calculating the gains from trade GT_d for any country d requires estimates of all of the model's exogenous features. These features include the observed inputs (tariffs $\{t\}$ and endowments $\{L\}$). They also include the model's unobserved parameters, such as the demand elasticity σ and bilateral technology shifters $\{A\}$ that apply to both competitive and monopolistically competitive models, plus the additional parameters for bilateral fixed costs $\{f\}$ and entry costs $\{f_E\}$ that apply in the case of the monopolistic competition model. Finally, and perhaps most fundamentally of all, the model's set of exogenous features includes the global distribution of productivities F that governs comparative advantage, trade possibilities, and hence the gains from trade.

3.1 Gravity models

Both of the special cases described in Section 2—the [Eaton and Kortum \(2002\)](#) version of the competitive model and the [Chaney \(2008\)](#) version of the monopolistic competition model—are distinctive because they work with a particular version F^* of the global productivity distribution F such that bilateral trade flows between countries take the so-called gravity form. Following [Arkolakis et al. \(2012\)](#) we therefore refer to these as gravity models.

In the case of gravity models, estimation of GT_d follows three well-known steps. Indeed, as shown by [Arkolakis et al. \(2012\)](#), when these three steps are executed on the same dataset the resulting value of GT_d will be identical for the [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) models. We therefore discuss the general approach that applies to both of these models simultaneously here.

Step 1: Estimate trade elasticity from orthogonality assumption. The first step is to estimate the single unknown parameter inside F^* , referred to in both models as θ^* . This involves estimating the slope—the so-called trade elasticity—in the gravity equation. Gravity models predict that aggregate bilateral trade flows X_{od} will follow the gravity equation:

$$\ln X_{od} = \alpha_o + \alpha_d - \theta^* \ln(1 + t_{od}) + \varepsilon_{od}. \quad (17)$$

Here, α_o and α_d collect model terms—including endogenous ones such as the wage of the origin and the income of the destination—that are common to a given origin and destination, respectively; and ε_{od} collects all exogenous model terms that are bilateral, other than the tariffs t_{od} . For example, in the [Eaton and Kortum \(2002\)](#) case, we have

$\alpha_o = -\theta^* \ln w_o$, $\alpha_d = \ln E_d - \ln \gamma_d$, and $\varepsilon_{od} = \theta^* \ln(A_{od})$. The [Chaney \(2008\)](#) case is similar but with added considerations for the endogenous number of varieties and bilateral fixed costs: $\alpha_o = \ln N_o + (1 - \frac{\sigma}{\sigma-1}\theta^*) \ln w_o$ and $\varepsilon_{od} = \theta^* \ln A_{od} + (1 - \frac{\theta^*}{\sigma-1}) \ln f_{od}$.

A standard approach to estimating the parameter θ^* therefore involves using data on observed bilateral trade flows, which we denote by $\tilde{X}_{od}$, and ad-valorem bilateral tariffs in order to estimate equation (17) by OLS. One approach to partialling-out the endogenous terms α_o and α_d is to double-difference equation (17) with respect to sample means. Letting, for example, $\Delta_o y_{od} \equiv y_{od} - \frac{1}{D} \sum_d y_{od}$ for any variable y_{od} , this procedure amounts to an OLS regression of $\Delta_o \Delta_d \ln \tilde{X}_{od}$ on $\Delta_o \Delta_d \ln(1 + t_{od})$.¹³ The coefficient from this regression will provide an unbiased estimate of $-\theta^*$ when the following orthogonality condition holds:

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \varepsilon_{od} = 0. \quad (18)$$

In the case of the [Eaton and Kortum \(2002\)](#) model the orthogonality condition (18) requires the assumption that tariffs are orthogonal to bilateral productivity shifters in the following sense

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln A_{od} = 0. \quad (19)$$

Similarly, consistent estimation for the [Chaney \(2008\)](#) case follows under the assumption that bilateral fixed costs are also orthogonal to tariffs:

$$\sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln f_{od} = 0. \quad (20)$$

Step 2: Estimate other model parameters by matching all trade flows. Armed with an unbiased estimate of θ^* from step 1, the second step uses this to provide unbiased estimates of the remaining model parameters of interest. This begins with the assumption that the model predictions for each of the D^2 bilateral trade flows X_{od} correspond to those in the data, namely

$$\mathbb{E}_{F^*}[X_{od}(\omega)] = \tilde{X}_{od} \quad \text{for all } o \text{ and } d. \quad (21)$$

Then, each of the D^2 bilateral residuals in equation (17)—i.e. $\ln \tilde{X}_{od} + \theta^* \ln(1 + t_{od})$ —provides an unbiased estimate of each of the D^2 terms ε_{od} , up to two irrelevant o -specific and d -specific scale factors. These estimated values of ε_{od} then provide (along with θ^*) unbiased estimates of the remaining model parameters that are necessary for counterfactual simulations (such as computing GT_d). For example, in the [Eaton and Kortum \(2002\)](#)

¹³In cases with no missing observations this is algebraically identical to an OLS regression of $\ln X_{od}$ on $\ln(1 + t_{od})$ and origin and destination fixed effects.

model, since $\varepsilon_{od} = \theta^* \ln(A_{od})$, the translation from each bilateral residual to estimates of each bilateral productivity parameter A_{od} is immediate. Similarly, standard arguments in the [Chaney \(2008\)](#) case imply that the same combination of A_{od} and f_{od} that is identified from ε_{od} is sufficient for all equilibrium calculations. Finally, a remaining unknown model parameter in both cases is σ but in both models this turns out to only affect the scale of consumer welfare, so it cancels out of GT_d .

Step 3: Solve for counterfactual changes in estimated model. The final step is to solve for GT_d , as defined in equation (12), according to the estimated model.¹⁴ In the case of the [Eaton and Kortum \(2002\)](#) version of the gravity model, this amounts to solving the system of equations (6) and (7) for the factual equilibrium and (9) and (10) for the counterfactual one, each time with the expectations $\mathbb{E}_F[\cdot]$ evaluated at $F = F^*$ (and F^* being independent draws from a *Frechet*(1, θ^*) distribution at the estimated value of θ^* obtained from step 1). Similarly, in the [Chaney \(2008\)](#) case, step 3 requires solving equations (14)-(16) for the factual equilibrium, and their counterfactual equilibrium analogs, while setting the distribution F^* to be independent draws from *Pareto*(1, θ^*) at the estimated value of θ^* . Either way, as discussed previously, the [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) models are empirically isomorphic—so whenever steps 1 and 2 have been executed on the same trade and tariff data, these two models return the same value of GT_d .

3.2 Beyond gravity models

We have seen how estimation of GT_d works in the case of the [Eaton and Kortum \(2002\)](#) version of the Ricardian model and the [Chaney \(2008\)](#) version of the monopolistic competition model. That is, in either case, we have seen how to estimate GT_d for one particular version, denoted F^* , of the global productivity distribution F . This raises two natural questions: How could the three-step estimation procedure of Section 3.1 be repeated for alternative distributions $F \neq F^*$? And how would GT_d change as these alternative distributions are considered?

One direction would be to repeat the analogs of steps 1-3 for particular alternative distributions. For example, one could imagine working with the Ricardian case and positing that the global distribution F corresponds to independent draws from a log-normal marginal distribution, with location parameter zero and scale parameter s , that is common to all countries. This would no longer be a gravity model. But the assumption that

¹⁴The second and third steps of this procedure are often combined into an algebraically equivalent single step via the “exact-hat” algebra procedure used by [Dekle et al. \(2009\)](#).

log tariffs $\ln(1 + t_{od})$ are mean independent of the log bilateral productivity shifter $\ln A_{od}$ could nevertheless be incorporated, via a simulated method of moments procedure, to estimate s and all of the values of A_{od} (up to scale) as residuals. Having done that, the model could be solved—evaluating the expectations $\mathbb{E}_F[\cdot]$ by simulation, since they are not available in closed form for this choice of F —and GT_d could be calculated for this particular version of the global productivity distribution.

A concern with this sort of exercise is that it (unavoidably) starts with a particular choice of F , and that choice may be difficult to motivate. To circumvent this concern we remain agnostic about the choice of F . Instead, we describe a procedure for estimating sharp bounds on GT_d that hold for *every* distribution F that lies within a given “neighborhood” of the reference distribution F^* . This permits a systematic investigation of the question posed above: how would GT_d change at alternative values of $F \neq F^*$?

Neighborhoods of the F^* distribution To begin, we use a standard method for comparing two distributions from the statistics literature. In particular, the “ ϕ -divergence” $D(F||F^*) \geq 0$ of F from F^* is given by

$$D(F||F^*) \equiv \int_{\{z\} \in \mathbb{R}_D^+} \phi \left(\frac{dF(\{z\})}{dF^*(\{z\})} \right) dF^*(\{z\}), \quad (22)$$

for a suitable convex function $\phi(\cdot)$. The form in $D(F||F^*)$ applies the penalty function $\phi(\cdot)$ to deviations of F from F^* at each point $\{z\}$ in the support of F^* and then integrates over all such penalties, weighted according to the distribution F^* . Several conventional choices of $\phi(\cdot)$ are available, such as the Kullback-Leibler (KL) form of $\phi(x) \equiv x \ln x - x + 1$ and the χ^2 form of $\phi(x) \equiv (x - e)^2 / (2e) + (x - e) + 1$. The particular definition of divergence that we use below is the “hybrid” divergence, recommended by [Christensen and Connault \(2023\)](#) for cases like ours where F^* has fat tails; this uses KL divergence for all $x \leq e$ and χ^2 divergence for $x > e$. Based on this definition of divergence, we define the neighborhood $\mathcal{N}_\delta$ that is “near” F^* as the set of all distributions F for which $D(F||F^*) \leq \delta$.

There are three attractive features of ϕ -divergence for this purpose. First, the divergence does not depend on the choice of units for the random variable $z_o(\omega)$. Second, neighborhood size δ is invariant under any transformations of location and/or scale in F^* . This means that our choice to model F^* with a common scale parameter in all countries does not affect any of our conclusions, nor does the fact that we set this parameter to one. Finally, as $\delta \rightarrow \infty$, any distribution that is absolutely continuous with respect to F^* —that is, any distribution that can be obtained by reweighting the baseline distribution within its support—can be approximated arbitrarily well by distributions contained

within $\mathcal{N}_\delta$. Since the [Eaton and Kortum \(2002\)](#) version of F^* has full support over positive values, and the [Chaney \(2008\)](#) version has full support above a value (of one) that is set without loss since it is the Pareto distribution's scale parameter, this means that all continuous distributions are considered as $\delta \rightarrow \infty$. This result is useful in cases where analytical bounds on the distribution-agnostic gains from trade are unavailable.

Bounds on GT_d in the Ricardian case. We now seek to estimate sharp bounds on the size of country d 's gains from trade GT_d that are consistent with the standard three-step estimation, but done for all productivity distributions F within the neighborhood $\mathcal{N}_\delta$ of the F^* assumed by [Eaton and Kortum \(2002\)](#).

To do so, we proceed as follows. Consider the upper bound on GT_d within $\mathcal{N}_\delta$, which we denote by $\overline{GT}_d(\delta)$. This can be written as the solution to the program

$$\begin{aligned} \overline{GT}_d(\delta) \equiv & \max_{F \in \mathcal{N}_\delta, \{A\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}} 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}} \\ \text{subject to: } & \begin{cases} \text{equilibrium } \{w\} \text{ and } \{\gamma\} \text{ hold at } \{t\}, \\ \text{equilibrium } \{w'\} \text{ and } \{\gamma'\} \text{ hold at } \{t'\}, \\ \mathbb{E}_F[X_{od}(\omega)] = \tilde{X}_{od} \text{ for all } o \text{ and } d, \\ \text{and } \sum_o \sum_d \Delta_o \Delta_d \ln(1 + t_{od}) \times \Delta_o \Delta_d \ln A_{od} = 0. \end{cases} \end{aligned} \quad (23)$$

That is, estimating $\overline{GT}_d(\delta)$ amounts to finding the version of the Ricardian model that maximizes the drop in welfare for country d when it moves from the initial equilibrium to autarky as per equation (12), while ensuring that: (i) admissible distributions F lie within the neighborhood $\mathcal{N}_\delta$; (ii) factual wages $\{w\}$ and price indices $\{\gamma\}$ satisfy the factual equilibrium equations (1), (3), (4), (6) and (7); (iii) counterfactual wages $\{w'\}$ and price indices $\{\gamma'\}$ satisfy the counterfactual equilibrium equations (8)-(10); (iv) each model-implied bilateral trade flow in the initial equilibrium matches that in the data $\tilde{X}_{od}$; and (v) bilateral productivity shifters $\{A\}$ are orthogonal to tariffs in the sense of equation (19). The analogous lower-bound, which we denote by $\underline{GT}_d(\delta)$, involves an identical program to that in (23) but with the min operator instead.

While the program in (23) is straightforward to state, obtaining its solution is far from trivial. For any $\delta > 0$ the set of distributions in $\mathcal{N}_\delta$ to be considered is infinite, so conducting a search in the space of distributions F is clearly infeasible. Fortunately, however, [Christensen and Connault \(2023\)](#) have shown that a particular sub-problem of (23) has a dual that is feasible. In particular, their result refers to cases that involve maximiz-

ing, over all distributions F in $\mathcal{N}_\delta$, an objective function that can be written as (a known function of) an expectation over F subject to any number of constraints that can also be written as (known functions of) expectations over F . In such a case, [Christensen and Connault \(2023\)](#) establish that the dual of this infinite-dimensional program is concave and finite-dimensional, with dimensionality given by the number of constraints, and where all expectations in this dual program are expectations over F^* . Moreover, under some regularity conditions, the solution converges as $\delta \rightarrow \infty$ to the sharp nonparametric bound (i.e., the supremum over essentially all F that are absolutely continuous with respect to F^*).

The program in (23) does not immediately fit this desired form where the objective function and all constraints involve only expectations over F . For example, the choice set in (23) directly involves model parameters such as the productivity shifters, $\{A\}$. Similarly, the factual and counterfactual equilibrium objects $(\{w\}, \{\gamma\})$ and $(\{w'\}, \{\gamma'\})$ enter the objective function (12) and constraints in nonlinear ways that disrupt the pure expectations form. To circumvent these challenges, [Christensen and Connault \(2023\)](#) propose that one treat all such variables—not just primitives such as $\{A\}$, but also endogenous variables such as $\{w\}$, $\{w'\}$, $\{\gamma\}$ and $\{\gamma'\}$ in our context—as an expanded set of “parameters”, which we denote ψ . After doing so, the program in (23) can be split into two stages. First, for any given value of the parameter vector ψ , solve the maximization program over $F \in \mathcal{N}_\delta$ by converting it from an infeasible infinite-dimensional program into a feasible finite-dimensional program. Second, optimize over ψ in an outer loop.

We execute this nested two-step procedure below. But the large dimensionality of our problem, with $D^2 + 4D$ parameters in the outer loop, requires us to utilize a number of observations about dimensionality-reduction that apply to our problem. This is discussed in [Section 3.3](#).

Bounds on GT_d in the [Melitz \(2003\)](#) case. Computing bounds on GT_d in the monopolistically competitive model can be done in an analogous manner to that of the competitive model. In this case we seek to bound GT_d across all versions of the [Melitz \(2003\)](#) model, with productivity distributions F within $\mathcal{N}_\delta$ of the i.i.d. *Pareto*($1, \theta^*$) distribution F^* used in the benchmark gravity model case of [Chaney \(2008\)](#). Yet, as above, we do so while replicating the standard three-step estimation procedure, which ensures that all bilateral trade flow data points are matched by the model at any of the distributions F under consideration, as well as that bilateral productivity shifters and fixed costs are orthogonal to tariffs in the sense of equations (18) and (20).

For the sake of interpretability we impose two restrictions on the set of [Melitz \(2003\)](#)

models that we evaluate here. First, we require that the counterfactual of interest does not change the mass of entrants, i.e. $N_o = N'_o$ for all o . This choice is conservative for evaluating sensitivity around the Pareto distribution F^* because N_o is always fixed in the F^* model.¹⁵ Second, we require that all firms in country d produce in autarky—i.e., that the endogenous cutoff in autarky, $(\hat{z}_{dd})'$, is equal to the lower bound of the support of F (i.e. one). This is also conservative because, in the absence of such a restriction, the optimizer could always change the bilateral fixed cost parameters $\{f\}$ so as to mimic a change in the support of F without actually changing F (and hence without using up any of the slack in the divergence constraint). The constraint of $(\hat{z}_{dd})' = 1$ also ensures that the interpretation of δ is similar to that in the Ricardian model, in which all goods are always produced domestically in autarky.

The program for maximizing GT_d in the monopolistic competition model is similar to that in (23), but now written as

$$\begin{aligned} \overline{GT}_d(\delta) \equiv & \max_{F \in \mathcal{N}_\delta, \{A\}, \{f\}, \{f_E\}, \{w\}, \{w'\}, \{\gamma\}, \{\gamma'\}, \{N\}} 1 - \left(\frac{\gamma_d}{\gamma'_d} \right)^{\frac{1}{1-\sigma}} \\ \text{subject to: } & \begin{cases} \text{equilibrium } \{w\}, \{\gamma\} \text{ and } \{N\} \text{ hold at } \{t\}, \\ \text{equilibrium } \{w'\}, \{\gamma'\} \text{ and } \{N\} \text{ hold at } \{t'\}, \\ N_o \mathbb{E}_F[X_{od}(\omega)] = \tilde{X}_{od} \text{ for all } o \text{ and } d, \\ \sum_o \sum_d \Delta_o \Delta_d \ln A_{od} \times \Delta_o \Delta_d \ln(1 + t_{od}) = 0, \\ \sum_o \sum_d \Delta_o \Delta_d \ln f_{od} \times \Delta_o \Delta_d \ln(1 + t_{od}) = 0, \\ \text{and } (\hat{z}_{dd})' = 1. \end{cases} \end{aligned} \quad (24)$$

Relative to the Ricardian model, the new considerations of the monopolistically competitive model naturally require an augmented set of model equations, as laid out in Section 2.5. It also requires a larger set of model parameters, due to the addition of bilateral fixed costs of production $\{f\}$ and entry costs $\{f_E\}$. In addition, the equilibrium number of potentially producing firms $\{N\}$ must be solved for at every candidate distribution F under consideration alongside the factual and counterfactual market-clearing wages $\{w\}$ and $\{w'\}$ and price indices $\{\gamma\}$ and $\{\gamma'\}$.

We solve the program (24), and its lower-bound equivalent $\underline{GT}_d(\delta)$, using the same inner- and outer-loop method described for the Ricardian case. However, the outer loop over parameters ψ is now substantially higher-dimensional due to the addition of D^2 additional bilateral fixed costs $\{f\}$, D additional entry costs $\{f_E\}$, and D additional firm

¹⁵That is, under F^* we have $N_o = \frac{\sigma-1}{\sigma\theta} \frac{L_o}{f_{E_o}}$, which is a constant.

masses $\{N\}$ to the vector ψ . The dimensionality-reduction techniques we discuss next are especially important in this case.

3.3 Implementation Details

We implement this two-step optimization program using Knitro, where an inner loop solves the search over distributions for a given vector of parameters, and an outer loop searches over the vector of parameters. We compute expectations using Monte Carlo integration with 800,000 draws.

In both versions of the model, there are a large number of parameters in the outer loop (in the order of at least D^2), which would be very challenging for any solver once the number of countries D is realistic. However, we can substantially reduce the dimensionality by exploiting the fact that many of these parameters are irrelevant for computing gains from trade for d . For the Ricardian case, the parameters $A_{od'}$ for $d' \neq d$ do not appear in any equations relevant for determining GT_d , which means they can be omitted.¹⁶ The only exception is that they do appear in the moment restriction corresponding to the gravity equation, discussed in Section 3.2, so we instead implement this moment using the subset of observations where $d' = d$.¹⁷ The same logic allows us to omit $\{\gamma_o, \gamma'_o, w'_o\}_{o \neq d}$. Since baseline trade flows imply a unique (to scale) vector of wages, $\{w_o\}$, we can solve for these directly. We can also normalize $w_d = w'_d = 1$ as well as one element of $\{A_{od}\}_o$. Finally, we can omit equation (6) as it is directly implied by the full set of trade flow moments. This means we are left with a parameter vector that consists of $\{\gamma_d, \gamma'_d, \{A_{od}\}_{o \neq 1}\}$, which is of dimension $D + 1$.

The Melitz (2003) case is similar, and actually benefits from similar dimensionality-reduction even though in its original form it is higher-dimensional than the Ricardian model. In addition to omitting $A_{od'}$ for $d' \neq d$ and $o \neq d$, we can also omit $\{A_{od}\}_{o \neq d}$. This is feasible because the constraint $N_d \mathbb{E}_F[X_{dd}(\omega)] = \tilde{X}_{dd}$ —which can never be omitted as it is necessary to identify A_{dd} and thus γ'_d —has no direct dependence on other countries, except through d -level aggregate objects (such as γ_d and N_d) that are treated as parameters.¹⁸ Moreover, combining the zero-profit condition (14) and the market clearing

¹⁶We also then omit the corresponding trade share constraints, $\tilde{X}_{od} = \mathbb{E}_F[X_{od'}(\omega)]$ for $d' \neq d$. This does not affect the model's ability to replicate the baseline equilibrium, as for any combination of F and the other parameters, it is possible to find a vector $\{A_{od'}\}_{o, d' \neq d}$ that can rationalize the omitted trade shares. The dimensionality reduction comes from the observation that we do not need to know the values of these omitted A_{od} to compute our counterfactual of interest.

¹⁷As this removes the ability to include origin fixed effects, the moment is equivalent to an instrumental variables estimation of $\ln \tilde{X}_{od}$ on $\ln w_o(1 + t_{od})$, using $\ln(1 + t_{od})$ as an instrument.

¹⁸This is not possible in the Ricardian case because the constraint $\mathbb{E}_F[X_{dd}(\omega)] = \tilde{X}_{dd}$ contains all $A_{od'}$ for

condition (16) allows us to eliminate all $A_{dd'}$ for $d' \neq d$ as aggregate profits are zero and these variables are not directly needed to determine d' 's consumption. These observations imply it is sufficient to aggregate all other countries $d' \neq d$ and reduce the overall set of parameters to $\{\gamma_d, \gamma'_d, A_{dd}, f_{dd}, f_{Ed}, A_{dx}, f_{dx}\}$, where A_{dx} and f_{dx} are auxiliary parameters that represent the technology shifter and the fixed costs in the aggregated export market with trade flow $X_{dx} \equiv w_d L_d - X_{dd}$. As a result, this case has only 7 parameters in the outer loop.

3.4 Data

Our analysis focuses on data from 2016. We obtain aggregate bilateral trade data $\tilde{X}_{od}$ from the OECD ICIO database and tariff data t_{od} from TRAINS, constructing a measure of the average bilateral tariffs by weighting 6-digit HS-level tariffs by their corresponding flows from the CEPII BACI database. In order to keep the dimensionality manageable, we work with a sample of countries that includes the 19 largest countries by total expenditure and aggregate all remaining countries into a Rest of World group. This results in $D = 20$.

4 Quantifying the sensitivity of quantitative trade models

We turn now to the following question: how sensitive are the conclusions that standard quantitative models make about the gains from trade to changes in the underlying distribution of productivities? For now, our focus lies on estimates of GT_d for the case where d is France, a country with import penetration that is representative among the countries in our sample.

4.1 Gains from trade in the gravity model

Workhorse quantitative trade models of the gravity form, such as [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#), predict how the welfare of a country d , such as France, would change if the country's tariffs were to rise to the point of autarky. These predictions about GT_d are made by employing the model estimation procedure detailed in Section 3.1 for a version of the model in which global production productivities F are drawn from the particular distribution F^* —the i.i.d. $Frechet(1, \theta^*)$ distribution in the case of [Eaton and Kortum \(2002\)](#) or the i.i.d. $Pareto(1, \theta^*)$ distribution in the case of [Chaney \(2008\)](#). Since the distribution F^* is, of course, a divergence $\delta = 0$ from itself, the point estimate of GT_d

$d' = d$ in the min function of equation (4).

obtained when assuming $F = F^*$ is the point at which our upper and lower bounds from Section 3.2 coincide, at the value $\overline{GT}_d(\delta = 0) = \underline{GT}_d(\delta = 0)$.

In addition, we know from the result of Arkolakis et al. (2012) that both the Eaton and Kortum (2002) and Chaney (2008) models will agree on the value of GT_d as long as they are estimated on the same underlying data in a manner that follows the estimation procedure of Section 3.1. That is, both models conclude that the gains from trade for country d follow the Arkolakis et al. (2012) formula

$$\overline{GT}_d(\delta = 0) = \underline{GT}_d(\delta = 0) = 1 - \lambda_{dd}^{\frac{1}{\theta^*}}, \quad (25)$$

where $\lambda_{dd} \equiv X_{dd}/w_d L_d$ is the share of d 's expenditure at baseline that it sources from itself. That is, these two models' predictions will agree if they are calibrated to the same baseline value of λ_{dd} , and if their estimated values of θ^* agree. And the models will agree on the estimated value of θ^* if their value of θ^* is estimated as (minus) the trade elasticity in the gravity equation (17) based on the same dataset.

We replicate standard estimation procedures in order to obtain an estimate of $\overline{GT}_d(\delta = 0) = \underline{GT}_d(\delta = 0)$ for France in 2016. First, the OLS estimate of θ^* in equation (17) is 6.8. Second, in the 2016 data of our application, France has a value of $\lambda_{dd} = 0.835$ (i.e. the country imports 16.5% of its expenditure). Putting these two numbers together, our estimate of France's gains from trade, according to the gravity model of either Eaton and Kortum (2002) or Chaney (2008), is $1 - (0.835)^{\frac{1}{6.8}} = 0.026$. In other words, according to the workhorse gravity model, aggregate welfare in France would drop by 2.6% if the country's trade policy were to retreat into autarky.

The remainder of our analysis is focused on exploring the sensitivity of this estimate to alternative distributions of global productivities.

4.2 Distribution-agnostic bounds on the gains from trade

A natural place to start when examining alternative productivity distributions is with the universe of all possible distributions. To this end, we now characterize the distribution-agnostic bounds on GT_d . Since, as remarked above, taking the $\delta \rightarrow \infty$ limit corresponds to the case where the neighborhood $\mathcal{N}_\delta$ around F^* is wide enough to either encompass or approximate arbitrarily well all distributions F that are absolutely continuous with respect to F^* , these upper and lower bounds can be thought of as $\overline{GT}_d(\delta = \infty)$ and $\underline{GT}_d(\delta = \infty)$, respectively. The following proposition characterizes the values of such bounds. We begin with the case of the competitive model but, as we show below, the bounds achieved

turn out to be identical in the competitive and monopolistically competitive models.

Proposition 1(a) (Bounds for competitive model). *Suppose that the baseline trade data satisfy, for country d of interest, the condition: $\lambda_{dd}/\lambda_{dn} \geq \lambda_{od}/\lambda_{on}$ for all $o, n \neq d$. Consider all possible productivity distributions F (i.e. $\delta = \infty$ and absolutely continuous with respect to F^*). Then the smallest and largest possible values for d 's gains from trade in the competitive model, as defined in program (23), are given by*

$$\underline{GT}_d(\delta = \infty) = 0 \quad \text{and} \quad \overline{GT}_d(\delta = \infty) = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (26)$$

The proof of this and all other propositions is in Appendix A. In all cases our proofs are constructive so they provide both a given GT_d bound (such as either of those in (38)) and an example of a distribution that achieves it. In this sense the agnostic bounds we provide are always sharp.

Before turning to the intuition behind Proposition 1(a), we comment on the data condition in its premise. This condition is easily verified in any given dataset and is satisfied for all countries (though not for the rest-of-world aggregator) in our application. This is the case because the diagonals of the trade share matrix (i.e. λ_{dd} , which describe countries' purchases from themselves) tend to be much larger than other elements in the matrix because most countries are not very open to trade.

Returning to the intuition, a first step is to note that one can combine the definition of λ_{dd} , the fact that autarky requires $\gamma'_d = \mathbb{E}_F[MC_{dd}(\omega)^{1-\sigma}]$, and the choice of numeraire ($w_d = w'_d = 1$), to write equation (12) as

$$GT_d = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}} \left(1 + \frac{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbb{1}\{d \neq \arg \min_o MC_{od}(\omega)\}]}{\mathbb{E}_F[z_d(\omega)^{\sigma-1} \mathbb{1}\{d = \arg \min_o MC_{od}(\omega)\}]} \right)^{\frac{1}{\sigma-1}}. \quad (27)$$

Consider the ratio of expectations in (27). Since productivity $z_o(\omega)$ is weakly positive, the smallest value this ratio can take is zero, in which case we obtain the upper bound of $1 - \lambda_{dd}^{\frac{1}{\sigma-1}}$ in Proposition 1(a). This happens when the domestic productivity on all goods that d 's consumers buy from abroad at baseline is zero, which implies that country d is unable to produce any of the goods it imports at baseline. This suggests that a key feature of any distribution F that can achieve the upper bound is that it puts a mass point with zero productivity for the goods that d imports at baseline.

A remaining challenge is to find such a distribution that is capable of matching the constraints in program (23)—generating an equilibrium that matches all baseline trade flow values and the orthogonality condition (19). The core content of Proposition 1(a) is

to show formally that as $\delta \rightarrow \infty$ the set of distributions F is flexible enough—through both its marginals and its dependence structure—to achieve this. Our specific construction starts with the distribution F^* which, by construction, matches all required constraints. We then remove from this distribution all the mass from the goods that d imports to zero productivity and then rescale it in a way that continues to satisfy all constraints. As we show, this is only possible when the data condition in the premise of Proposition 1(a) is satisfied.

Turning to the lower bound in Proposition 1(a), an equivalent way of writing (27) is

$$GT_d = 1 - \left(\frac{\mathbb{E}_F[MC_{dd}(\omega)^{1-\sigma}]}{\mathbb{E}_F[(\min_o MC_{od}(\omega))^{1-\sigma}]} \right)^{\frac{1}{\sigma-1}}. \quad (28)$$

Seen this way, GT_d would achieve the minimal value of zero if the distribution of productivities F could be such that the expected marginal cost from domestic production is the same, on average, as that from the cheapest global seller to d . This occurs in the opposite type of scenario from the upper bound case: it requires that country d can produce all imported goods at exactly the same tariff-inclusive prices as occur at baseline. Then baseline imports merely reflect indifference, so shutting down trade need not raise the price index at all and gains from trade can be zero. Again, we construct an example that achieves this by starting from F^* , adding mass to the segments necessary for such indifference to hold, and then rescaling subject to the constraints in (23). It turns out that this is again only possible when the data condition in the premise of Proposition 1(a) is satisfied.

Having characterized distribution-agnostic bounds for the competitive model, the case of the monopolistic competition model proceeds very similarly.

Proposition 1(b) (Bounds for monopolistic competition model). *Suppose that the baseline tariff data satisfy the condition that there exists at least one pair of countries, k_1 and k_2 , with $k_1 \neq k_2$, such that $\Delta_o \Delta_d \ln(1 + t_{k_1 k_2}) > 0$. Consider all possible productivity distributions F (i.e. $\delta = \infty$ and absolutely continuous with respect to F^*). Then the smallest and largest possible values for d 's gains from trade in the monopolistic competition model, as defined in program (24), are given by*

$$\underline{GT}_d(\delta = \infty) = 0 \quad \text{and} \quad \overline{GT}_d(\delta = \infty) = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (29)$$

Just as with the Ricardian case above, the constructions we offer for achieving these bounds require that a data condition—stated in the premise of Proposition 1(b)—is satisfied. In this case the condition involves only tariff data and is extremely mild. Indeed, it holds generically whenever $D \geq 3$, as in our application.

The intuition for the monopolistically competitive model is similar to that seen above

and, as before, constructive arguments are contained in Appendix A. The key distinction relative to the competitive model is that each good is now a distinct variety globally. Country d 's gains from trade can therefore achieve the bounds in Proposition 1(b) by adjusting the distribution of productivities among the potential producers in d that will begin production in autarky so that they are either highly productive (which minimizes GT_d) or unproductive (which maximizes it). Once again, this construction is trivial apart from the need to ensure that such a distribution can satisfy the constraints in program (24). Our contribution is to establish that the set of distributions F is flexible enough to do so while matching the constraints, provided that the stated tariff data condition holds.

We turn now to calculating the upper bound in Propositions 1(a) and 1(b) for France in our dataset. Like the Arkolakis et al. (2012) formula for GT_d under $\delta = 0$, this expression for the maximum possible value of GT_d depends on the baseline value of λ_{dd} , discussed above. But rather than depending on the trade elasticity θ^* , the upper bound $\overline{GT}_d(\delta = \infty)$ depends on σ , the elasticity of substitution across products. Perhaps surprisingly, the value of σ does not enter the $GT_d(\delta = 0)$ formula under F^* at all, so the standard estimation template for GT_d in gravity models does not deliver (or require) an estimate of σ . But σ does matter for counterfactuals under essentially all other possible distributions $F \neq F^*$. We therefore work with the value of $\sigma = 2.5$ and explore robustness to this value in ongoing work.

Using this value of σ in the formula $\overline{GT}_d(\delta = \infty) = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}$ implies that the maximal value that France's gains from trade can take, across all possible production heterogeneity distributions F , is 11.3% (i.e. $1 - (0.835)^{\frac{1}{2.5-1}}$). This value is 4.4 times larger than that of 2.6% predicted by the workhorse model under F^* . Similarly, the lower bound $\underline{GT}_d(\delta = \infty) = 0$ in Propositions 1(a) and 1(b) is meaningfully different from that implied at F^* . Evidently, if one were completely agnostic about the model's underlying production heterogeneity then a very wide range of predictions about GT_d are possible.

4.3 Interpreting measures of divergence δ

One reaction to the wide bounds on GT_d across all distributions (i.e. for $\delta = \infty$) discussed above is that complete agnosticism is too nihilistic. We therefore re-evaluate these bounds for a smaller set of productivity distributions by narrowing the neighborhood size $\mathcal{N}_\delta$ such that δ is finite (but still greater than zero). Before exploring such neighborhoods, we first pause to anticipate the question of how a divergence of any given value, such as δ , can be interpreted.

To provide one benchmark, consider the sensitivity analysis conducted for the case of

gravity models when the value of the trade elasticity is adjusted, as is common practice among users of quantitative trade models. Begin with the [Eaton and Kortum \(2002\)](#) case, where the global productivity distribution F^* implies that $z_o(\omega)$ is distributed, across o and ω , such that each country o draws its own $z_o(\omega)$ independently from the distribution $\text{Frechet}(1, \theta^*)$. Now consider changing this distribution to an alternative, denoted $F^{\text{alt.}}(\theta)$, that remains an [Eaton and Kortum \(2002\)](#) gravity model but with a different trade elasticity, θ . That is, $F^{\text{alt.}}(\theta)$ is analogous to F^* but based on $\text{Frechet}(1, \theta)$ at some $\theta \neq \theta^*$. Because this distribution $F^{\text{alt.}}(\theta)$ differs from F^* we can calculate the resulting divergence, $D(F^{\text{alt.}}(\theta) || F^*)$.

The red line in Figure 1 plots the value of $D(F^{\text{alt.}}(\theta) || F^*)$ against θ for this Frechet case estimated in Section 4.1.¹⁹ That is, F^* corresponds to $D = 20$ countries taking independent draws from $\text{Frechet}(1, \theta^* = 6.8)$. Similarly, the blue line reports an analogous exercise for the case of [Chaney \(2008\)](#), where F^* is based on a $\text{Pareto}(1, \theta^* = 6.8)$ distribution. Naturally, in both cases the divergence is zero at $\theta = \theta^* = 6.8$, and the divergence is strictly greater than zero for all other values of $\theta \neq \theta^*$. Our interest lies in the extent to which the divergence grows as θ differs increasingly from θ^* .

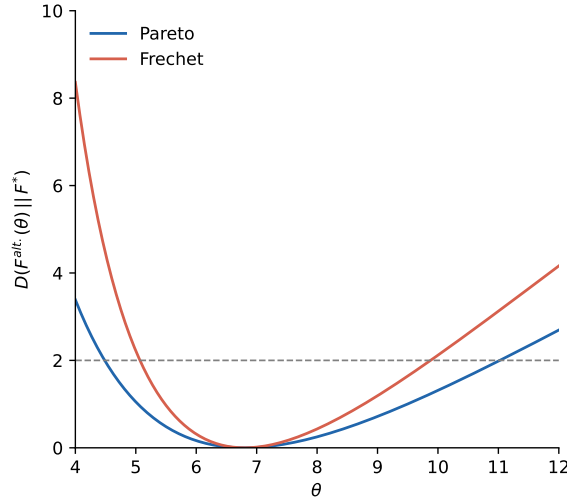


Figure 1: Examples of divergence between distributions

For example, we see that divergence $D(F^{\text{alt.}}(\theta) || F^*)$ below the value of 2 corresponds to trade elasticities θ that span the range $\theta = [5.2, 9.8]$ in the Frechet case and $\theta = [4.5, 11]$ in the Pareto case. These windows seem well within the range of conventional sensitivity

¹⁹The calculations in Figure 1 use the KL divergence rather than the Hybrid divergence used in all further analysis below. This is conservative (because KL divergence is always weakly smaller than Hybrid divergence) and avoids computation of high-dimensional numerical integrals (since KL divergence is available in closed form for the two cases calculated here).

analysis that is already a key part of quantitative trade modeling.²⁰ We therefore view a neighborhood such as $\mathcal{N}_\delta$ at $\delta = 2$ —that is, the set of *all* distributions F that lie within $D(F||F^*) \leq 2$ of F^* —to be of substantial interest when conducting sensitivity analysis.

4.4 Sensitivity of the gains from trade to alternative productivity distributions

We now turn to the main goal of our analysis, which is to estimate bounds on counterfactual predictions about the gains from trade that hold for a varying set of neighborhoods $\mathcal{N}_\delta$ centered around the workhorse gravity model.

As described in Section 3.2, for any given value of δ , we calculate the upper and lower bounds on country d 's gains from trade, $\overline{GT}_d(\delta)$ and $\underline{GT}_d(\delta)$ respectively, that would be possible across all productivity distributions with divergence less than δ from the gravity model distribution F^* . And we do so while always ensuring that, given any distribution under consideration, it is fed into an otherwise standard equilibrium model that exactly matches all trade flow data points at the factual equilibrium. We focus on the case where $d = \text{France}$.

Figure 2 plots our baseline results. For purposes of comparison, the dashed orange line displays the value of $\overline{GT}_d(\delta = 0)$, the extent of gains from trade in the workhorse model (i.e. $\delta = 0$); as discussed in Section 4.1, we estimate this to be $\overline{GT}_d(\delta = 0) = 0.026$. In addition, the dashed black lines report the distribution-agnostic upper and lower bounds, $\overline{GT}_d(\delta = \infty)$ and $\underline{GT}_d(\delta = \infty)$ respectively, that are possible across all distributions; as discussed in Section 4.2, we estimate these bounds to be $\overline{GT}_d(\delta = \infty) = 0.113$ and $\underline{GT}_d(\delta = \infty) = 0$. On top of these lines we add the values of $\overline{GT}_d(\delta)$ and $\underline{GT}_d(\delta)$ as a function of δ for the case of the Ricardian model (in solid blue) and the Melitz (2003) model (in solid red), each computed for five values of δ at each bound. As expected, the bounds begin at the standard gravity model estimate of $\overline{GT}_d = 0.026$ when δ is small and then approach the distribution-agnostic bounds $\overline{GT}_d(\delta = \infty)$ and $\underline{GT}_d(\delta = \infty)$ as δ grows larger.

Of greater interest is the *rate* at which the bounds diverge as δ is increased, starting at zero. This encapsulates a notion of sensitivity to alternative productivity distributions, and Figure 2 reveals a striking amount of sensitivity. For example, at $\delta = 1$ the trade data support Ricardian models with gains from trade that are four times larger than in the gravity model, as well as Melitz (2003) models that are also four times larger. And at $\delta =$

²⁰For example, Eaton and Kortum (2002) deploy three procedures for estimating θ^* and arrive at values ranging from 3.60 to 12.86.

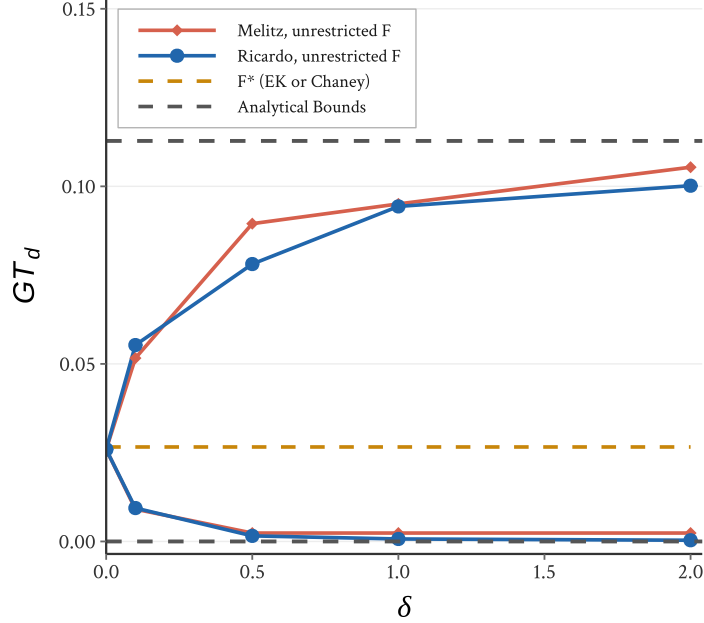


Figure 2: Bounds on gains from trade (for France) across alternative productivity distributions

2—a value of divergence that, as discussed in Section 4.3, we view as corresponding to a conventional exploration of robustness in gravity models—the upper bounds in Figure 2 are very close (88% of the way for Ricardo and 92% for Melitz (2003)) to the distribution-agnostic maximal upper bound that, as discussed above, is itself about 4.4 times larger than in the gravity model. Evidently, a neighborhood of divergence within $\delta = 2$ of the gravity model F^* comes close to spanning the space of relevant distributions that matter for gains from trade in these models. That is, it includes distributions that lead to viable general equilibrium trade models of either perfect or monopolistic competition, that are completely capable of matching all data on trade flows, that satisfy standard orthogonality conditions, and that can approximately achieve the maximal and minimal gains from trade that are possible across all distributions in these trade models.

5 Restricting Productivity Distributions

We have seen in Figure 2 how the estimates that workhorse quantitative trade models provide about gains from trade are highly sensitive to the model’s assumption about the underlying productivity distribution F . That is, even for the neighborhood of distributions that are “close” to the benchmark of F^* , gains from trade can be many times smaller or larger than they are in the model that assumes F^* . These findings beg the question:

how much narrower would the bounds on GT_d be if one were to consider only a restricted class of distributions F ?

In this section we provide a set of answers to this question. Of course, there are many ways to restrict the types of distributions under consideration.²¹ We focus on a set of high-level characteristics that are assumed in F^* and examine how the bounds on GT_d change as we impose some of these characteristics, one by one.

The global productivity distributions F^* used by [Eaton and Kortum \(2002\)](#) in the competitive model and [Chaney \(2008\)](#) in the monopolistically competitive model embody three core features. First, each country takes independent draws—that is, there is no cross-country dependence or correlation across their draws. Second, every country draws from the same marginal distribution. And third, the particular marginal distribution that all countries draw from is $Frechet(1, \theta^*)$ in the [Eaton and Kortum \(2002\)](#) case and $Pareto(1, \theta^*)$ in the [Chaney \(2008\)](#) case.

So far, our investigations have searched over the neighborhood $\mathcal{N}_\delta$ for distributions that, while always generating trade models that match the data, are otherwise unrestricted. This means that the distributions $F \in \mathcal{N}_\delta$ considered are relaxing some or all of the three high-level characteristics held by F^* as they move throughout the less restricted space of distributions. In this section we impose constraints on this flexibility—subsets of the three high-level conditions embodied in F^* —in order to learn about which conditions are more restrictive in the economic sense of constraining the gains from trade.

5.1 Restrictions: competition vs. monopolistic competition

One starting observation concerns a broad distinction between the competitive Ricardian and monopolistically competitive [Melitz \(2003\)](#) models. The competitive model involves head-to-head competition, in every destination market d , over the worldwide producers of each good ω who are based in each of the D origin countries. When country d moves from its factual trading equilibrium to the counterfactual state of autarky, the supplier of every imported good will shift from the foreign supplier (who had won this head-to-head competition in the factual equilibrium) to a domestic supplier. The gains from trade therefore depend on the size of the foreign supplier's victory. The probabilistic properties of this victory depend on country d 's marginal productivity distribution and the correlation of d 's draws with the best of the $D - 1$ foreign draws, a distribution that itself depends on those $D - 1$ foreign countries' marginal distributions and their dependence. In short, the

²¹Recall that the divergence between distributions is invariant to a change in the units of measurement in either distribution. So one type of restriction that is irrelevant to our conclusions is the pure location (or units of account) of the distributions we consider.

gains in the Ricardian model clearly depend on on properties of domestic marginals, foreign marginals, and the dependence among them. We explore these components below.

On the other hand, consider the [Melitz \(2003\)](#) model. In this case, competition is not head-to-head for any good; to the contrary, every firm pays an entry cost in order to be the unique global monopolist in its own product. When country d moves to autarky it will lose access to all foreign imports and seek to replace them with domestic alternatives. But the lack of head-to-head competition means that what matters for the gains is how well these newly produced domestic varieties can replace the aggregate contribution of the now-unavailable foreign imports, which is purely a function of the domestic distribution. That is, conditional on knowing the marginal distribution of productivities in country d the additional properties of F —such as the $D - 1$ foreign countries' marginal distributions or the cross-country dependence of productivity draws—has no direct bearing on GT_d . Given these points, there is little to be learned by imposing high-level restrictions on the Melitz model. All results in the following subsections therefore focus on the Ricardian case alone.

5.2 Imposing independence and zero correlation

As discussed, the F^* used by the [Eaton and Kortum \(2002\)](#) version of the Ricardian model assumes that, for any good ω , countries o take their productivity draws $z_o(\omega)$ independently from one another. By contrast, the distributions F that we have considered so far have allowed for arbitrary forms of cross-country dependence. We now seek to narrow this gap between F^* and F by restricting the set of distributions F considered such that any admissible F must satisfy cross-country independence just like F^* does. In this case the only distinction between F^* and F that remains is that due to the particular marginal distributions that each country draws from.

Distribution-agnostic bounds. It is possible to extend the upper-bound argument referring to all distributions (Proposition [1\(a\)](#)) to the case of distributions F featuring independence across countries. This is perhaps surprising given that the construction used for all-distributions case exploited a strong form of dependence across countries. But as the following proposition demonstrates, the space of distributions F is flexible enough to allow (provided a stronger data condition holds) the same bound to be achieved without exploiting any cross-country dependence.

Proposition 2 (Bounds for competitive model with independence). *Suppose that the base-line tariff and trade data satisfy the following condition: letting Ξ denote a square matrix composed*

of elements Ξ_{ij} that are zero apart from the d th row with elements given by $\frac{1}{1-\sigma} \ln \frac{\lambda_{dj}}{1-\lambda_{dj}}$, we have $\text{Var}(\Delta_i \Delta_j \ln(1 + t_{ij})) > \text{Cov}(\Delta_i \Delta_j \Xi_{ij}, \Delta_i \Delta_j \ln(1 + t_{ij}))$. Consider all possible productivity distributions F under which countries draw their productivities independently. Then the largest possible value for country d 's gains from trade in the competitive model, as defined in program (23), is given by

$$\overline{GT}_d^{\text{indep.}} = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (30)$$

Further, this value is approached arbitrarily closely by a sequence of distributions that have Frechet marginals.

Like those seen earlier, this proposition presumes a data condition that can be easily checked on any dataset and is verified for all countries in our application. The proof of Proposition 2 provides a construction of F that involves countries drawing independently from a distribution with (non-identical) Frechet marginals. The marginal for country d has a shape parameter that approaches $\sigma - 1$, in which case that country's production is dominated by a small mass of highly productive goods, such that in autarky the contribution of other goods is negligible, i.e., the denominator in (27) becomes arbitrarily large. All other countries, apart from d , have a common marginal distribution that we set to be Frechet with shape parameter increasing from θ^* until the constraints in program (23) are satisfied. As we show, this is always possible as long as the data condition in the premise of Proposition 2 holds. Intuitively, the construction balances the low model-implied trade elasticity in d with a higher model-implied trade elasticity in other countries, and the data condition rules out a case where d 's exports are so (empirically) elastic to tariffs that the A_{od} required to rationalize the data as $\theta \rightarrow \sigma - 1$ are simply too negatively correlated with tariffs to be balanced by those of other countries.

Bounds in the neighborhood of F^* . Proposition 2 suggests that when one considers all distributions it is essentially without loss to impose independence on the competitive model. However, as the construction provided suggests, this may be a limiting result that may not be very accurate in any neighborhood that is "near" to F^* . We therefore turn to an exploration of such neighborhoods in the case where they are restricted to include only distributions $F \in \mathcal{N}_\delta$ that satisfy independence.

Recall that two jointly distributed random variables are independent if and only if it is the case that their joint CDF is equal to the product of their two marginal CDFs. We approximate this restriction about CDFs by using a finite number of expectations at L^I specific quantiles of the relevant CDFs. That is, we say that F satisfies approximate

independence when

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] = \mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}]\mathbb{E}_F[\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}], \quad (31)$$

holds for all pairs of countries $o \neq o'$ and all L^I quantiles l , and where $\bar{z}_l$ denotes the value of the l th quantile of a distribution of interest.²² Since this constraint imposes the necessary and sufficient condition for independence at a finite number of points in the distribution, by letting L^I grow we can approximate increasingly well the case of true independence.²³

We impose this form of independence on F by adding the $D(D-1)L^I$ constraints embodied in (31) to the program (23) when computing the upper bound $\overline{GT}_d(\delta)$ and the lower bound $\underline{GT}_d(\delta)$. This continues to search over all distributions F that deliver Ricardian models that are consistent with trade data and the standard orthogonality condition, but with the space of such distributions now confined to those that feature (approximate) independence of productivity draws across countries.

Figure 3 displays the results from implementing this independence restriction: blue lines show the unrestricted bounds from Figure 2 and the dark green lines show those that impose independence. In the case of the lower bound $\underline{GT}_d(\delta)$, these lines are essentially coincident, implying that the bounds are unperturbed by the requirement that F features cross-country independence. On the other hand, the upper bound $\overline{GT}_d(\delta)$ with independence is somewhat lower than that facing no restrictions. For example, at the $\delta = 2$ level, $\overline{GT}_d(\delta)$ is 0.086 when F must exhibit cross-country independence and 0.1 when F is unrestricted. This implies that the maximum possible gains from trade in France would be about 18% (i.e. $1 - (0.086 - 0.026)/(0.1 - 0.026)$) less sensitive to the underlying productivity distribution F if it were believed that countries draw their technologies independently. As anticipated, even though Propositions 1(a) and 2 imply that imposing an independence restriction has no “bite” at $\delta \rightarrow \infty$, the bite is non-negligible at $\delta = 2$.

Stopping short of full independence, one may wish to impose weaker restrictions on the dependence of productivity draws across countries. For example, a condition of zero correlation can be obtained by imposing the far simpler version of (31) based on first

²²In practice, we use the common marginals of F^* , i.e. $\text{Frechet}(1, \theta^*)$, for this purpose. This has the advantage of using quantile values that are spaced out in the same proportion to their distribution under F^* .

²³In practice, we set $L^I = 5$ due to the large number of pairwise combinations in (31).

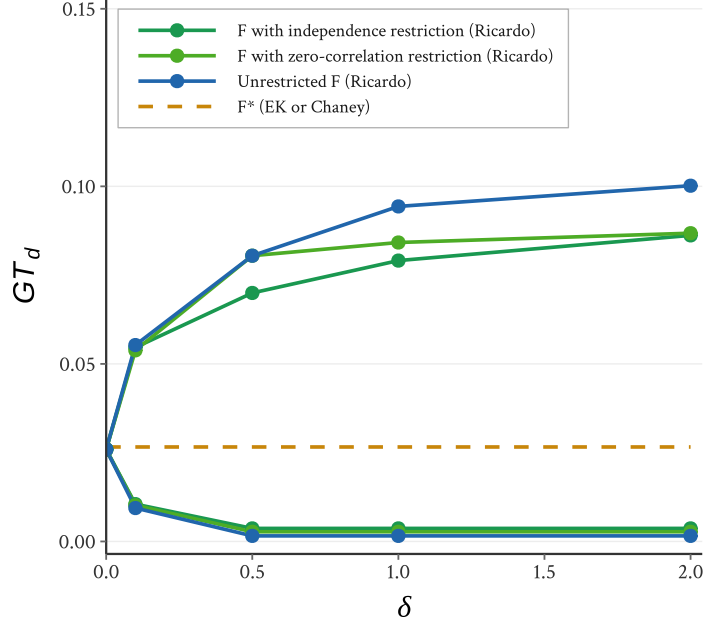


Figure 3: Ricardian gains from trade and cross-country dependence

moments alone:

$$\mathbb{E}_F[z_o(\omega)z_{o'}(\omega)] = \mathbb{E}_F[z_o(\omega)]\mathbb{E}_F[z_{o'}(\omega)] \quad \text{for all } o \neq o'. \quad (32)$$

We report on this zero-correlation case in the light green line shown in Figure 3. Naturally, since this restriction is weaker than that of full independence, the zero-correlation restriction bounds are weakly wider than the approximate independence restriction bounds (dark green lines). But the lines are close, especially for δ above 0.5. There is therefore not much of a distinction to be had from more complicated forms of dependence than linear correlation—a finding that is perhaps unsurprising, at least for high δ , given that even the strong restriction of full independence can achieve the maximal GT_d bound (Proposition 2).

These results are also informative for the prior literature that has relaxed the independence assumption in Eaton and Kortum (2002). For example, a special case of the analysis in Lind and Ramondo (2023) maintains the use of Frechet marginals but allows for non-zero amounts of correlation. Figure 3 illustrates an analogous idea but in the opposite direction: start with all distributions (blue lines) and then impose restrictions so as to keep only those with zero correlation (light green lines). We learn that this can reduce the maximal gains from trade, but not by much (only by 18% at $\delta = 2$). And it has essentially no effect on the minimal gains from trade.

5.3 Imposing common marginals

The second high-level restriction imposed in the [Eaton and Kortum \(2002\)](#) choice of F^* is that all countries o draw their productivities $z_o(\omega)$ from a common marginal distribution. Of course, the actual distribution F^* involves not only common marginals but specific marginals given by $\text{Frechet}(1, \theta^*)$. But the feature of commonality may derive from primitive assumptions on where technologies derive from, or simply from a desire for parsimony. So it is interesting to evaluate how the case of common (but flexible) marginals alone can shape gains from trade in data-consistent Ricardian models.

Distribution-agnostic bounds. We begin again with an extension of Proposition [1\(a\)](#) to the case where admissible distributions F must feature common marginals. In fact, the following proposition goes further and imposes the additional restriction that those common marginals must all be $\text{Frechet}(1, \theta)$ —though potentially with a shape parameter θ that differs from that value θ^* used in F^* .

Proposition 3 (Bounds for competitive model with common marginal distributions). *Suppose that the baseline tariff and trade data satisfy the data condition $\mathcal{C}^P$ stated in Appendix [A.4](#). Consider all possible productivity distributions F under which countries draw their productivities from a common marginal distribution but with potential dependence across countries. Then the largest possible value for country d 's gains from trade in the competitive model, as defined in program [\(23\)](#), is given by*

$$\overline{GT}_d^{\text{CM}} = 1 - \lambda_{dd}^{\frac{1}{\sigma-1}}. \quad (33)$$

Further, this bound is approached arbitrarily closely by a distribution with $\text{Frechet}(1, \theta)$ marginals.

The data condition behind this proposition is more involved than those seen earlier. This is unsurprising, given that stronger restrictions have been placed on the possible distributions that must satisfy all constraints despite achieving the same bound. While we relegate the exact condition to the Appendix, the rough idea is to partition all countries (other than d) into K groups in order to ensure that the sum across groups of certain within-group covariances (between tariffs and certain trade data summaries) is smaller than the overall variance of the tariff data. We find that this condition is satisfied for our application even when only $K = 2$ groups are considered.

Armed with this data condition, the proof of Proposition [3](#) constructs a sequence of common Frechet distributions with shape parameter θ that approaches $\sigma - 1$, while using the dependence structure to ensure that the orthogonality condition continues to hold. A common shape parameter below θ^* would normally imply that unobserved productivity must be negatively correlated with tariffs, as the model needs to explain why trade is

more elastic to tariffs than θ suggests. However, the dependence structure provides a separate lever to explain this: it can make draws correlated within blocks of countries, which (all else equal) decreases trade and thus behaves like a negative productivity shifter. The data condition ensures that constructing these blocks of countries with high intra-block correlation is sufficient to fully compensate for the changes in θ .

Bounds in the neighborhood of F^* . Turning to the case of finite δ , like in the previous subsection we approximate a restriction about (marginal) CDFs by using a finite number of expectations at L^{CM} quantiles l . All countries' marginal distributions are approximately common if the following condition holds

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{z}_l\}] = \mathbb{E}_F[\mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] \text{ for all } l \leq L^{CM} \text{ and } o' \neq o. \quad (34)$$

That is, common marginals across countries requires that the expected mass of one country's (e.g. o 's) distribution below any given point $\bar{z}_l$ is the same as that for any other country o' .²⁴ The gains from trade bounds, such as $\overline{GT}_d(\delta)$, among all distributions F for which all countries share the same marginal distribution can then be approximated by adding the $(D-1)L^{CM}$ constraints embodied in (34) to the program (23).

While this method works arbitrarily well for large L^{CM} , one challenge in practice occurs in regards to the likelihood of the highest-productivity draws, which are particularly important for generating gains from trade. Since gains from trade depend on the price index, which is a transformation of the $(1-\sigma)$ th moment of the price draws that consumers face, we mitigate the risks of inadequate upper-tail approximations by further imposing the additional $(D-1)L^{CM}$ constraints

$$\mathbb{E}_F[z_o(\omega)^{(1-\sigma)} \mathbb{1}\{z_o(\omega) < \bar{z}_l\}] = \mathbb{E}_F[z_{o'}(\omega)^{(1-\sigma)} \mathbb{1}\{z_{o'}(\omega) < \bar{z}_l\}] \text{ for all } l \leq L^{CM} \text{ and } o' \neq o. \quad (35)$$

This ensures that the draws beneath each quantile have the same $(1-\sigma)$ th moment.

The orange lines in Figure 4 report the results from this exercise. (Again, the blue lines display the unrestricted baseline case for comparison.) The message from this figure is that imposing the common marginals constraint does not do much to reduce the sensitivity of quantitative Ricardian models—not just at large δ as in Proposition 3, but even at small values of δ such as one or 2. In particular, the lower bound of zero is un-

²⁴We again choose the points $\bar{z}_l$ as the L^{CM} quantiles of the $Frechet(1, \theta^*)$ distribution. For any given L^{CM} , the common marginals restriction in (34) requires far fewer restrictions than the pairwise restrictions required for independence in (31), so it is feasible to use a larger number of quantiles. We therefore set $L^{CM} = 50$ here.

affected by this constraint and the upper bound is reduced, but only by about 11% (i.e. $1 - (0.092 - 0.026) / (0.1 - 0.026)$) at a value such as $\delta = 2$.

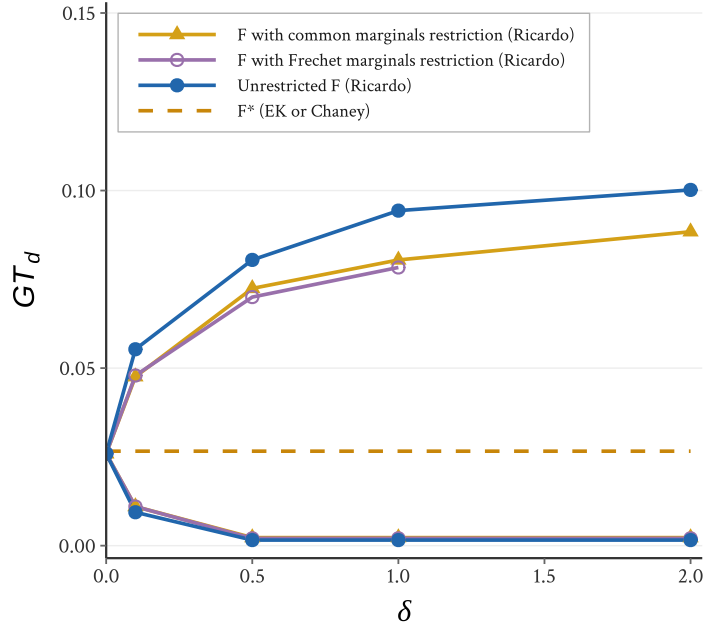


Figure 4: Ricardian gains from trade and countries' marginal distributions

5.4 Imposing common $Frechet(1, \theta^*)$ marginals

The previous restriction on F requires that all countries share the same marginal distribution, but it remains agnostic about what that marginal distribution is. A tighter restriction of interest is that the common marginal distribution takes a particular form, such as the $Frechet(1, \theta^*)$ distribution, as in the F^* used by [Eaton and Kortum \(2002\)](#).²⁵ We do this next. It is worth noting that in this case the only difference between F and F^* is that F remains agnostic about the cross-country dependence structure, whereas F^* assumes independence.

To see how this can be done, let $H(\cdot)$ denote the CDF of the inverse of the distribution $Frechet(1, \theta^*)$. Divide the unit-interval support of $H(\cdot)$ into $\bar{H}$ discrete and evenly-spaced points indexed by h . Finally, let $\bar{H}_h$ denote the value of the h th quantile of $H(\cdot)$. Then we can ensure that the marginal of the distribution of productivity draws $z_o(\omega)$ in any country o has approximately the same marginal distribution as $Frechet(1, \theta^*)$, as desired,

²⁵This is a distinct exercise from that in Proposition 2. Here we constrain the common marginal to be $Frechet(1, \theta^*)$, with θ^* set to the estimated value of 6.8; the construction in Proposition 2 instead uses common $Frechet(1, \theta)$ marginals with a flexible $\theta > \sigma - 1$.

by matching each of these $\bar{H}$ quantile values, as long as $\bar{H}$ is large enough for this approximation to be accurate. Formally, this amounts to imposing the constraint

$$\mathbb{E}_F[\mathbb{1}\{z_o(\omega) < \bar{H}_h\}] = H(\bar{H}_h) \quad \text{for all } o \text{ and all } h \leq \bar{H}. \quad (36)$$

In addition, for reasons discussed in Section 5.3, we ensure that the $(1 - \sigma)$ th moment of each quantile of every country's distribution is the same as that under $Exp(1)$:

$$\mathbb{E}_F[z_o(\omega)^{1-\sigma} \mathbb{1}\{z_o(\omega) < \bar{H}_h\}] = \mathbb{E}_{F^*}[z_o(\omega)^{1-\sigma} \mathbb{1}\{z_o(\omega) < \bar{H}_h\}] \quad \text{for all } o \text{ and all } h \leq \bar{H}. \quad (37)$$

Adding the $2D\bar{H}$ constraints in (36) and (37) to the program (23) ensures that bounds on GT_d at any δ only consider distributions F that deviate from F^* in terms of the cross-country dependence of their technology draws and not at all (at least approximately) in terms of the Frechet marginals that all countries share.²⁶

We report on this case in Figure 4. The purple lines impose the new restriction of common $Frechet(1, \theta^*)$ marginals. Again, counterfactual sensitivity remains substantial, despite requiring that all countries have the same marginal distributions as they do in Eaton and Kortum (2002). Perhaps surprisingly, the purple lines (where all marginals are the same and Frechet) are very close to the orange lines (where all marginals are the same but can be *any* marginal distribution). So, at least when dependence is left flexible, the sensitivity of the Ricardian gravity model's predictions has little to do with the particular marginal distributions employed.

5.5 Imposing both independence and common marginals

The four restrictions we have introduced so far have all been imposed one at a time. But subsets of them can also be imposed simultaneously. One combination of interest is the case where F displays cross-country independence and all countries share the same marginal distributions, in which case the only remaining distinction relative to F^* is that the marginals are flexible rather than $Frechet(1, \theta^*)$. We do this by adding to the program (23) the stacked set of $D(D - 1)L^I = 1900$ constraints in (31) and the $2(D - 1)L^{CM} = 1900$ constraints in (34) and (35).

The results of this exercise are shown in Figure 5. In particular, the light blue lines report on the case where all permitted distributions F satisfy both (approximate) independence and common marginals. For comparison, the dark blue lines again display the unrestricted case and the orange lines display that with common marginals (but not inde-

²⁶In practice we set $\bar{H} = 50$.

pendence). Once again, the bounds conveyed by the light blue lines are still quite wide. Going from the unrestricted case to the doubly-restricted case of both independence and common marginals achieves, at $\delta = 2$ for example, a reduction in sensitivity of the upper bound by about 23.5% (i.e. $1 - (0.082 - 0.026) / (0.1 - 0.026)$). What is most surprising is that the combination of independence and common marginals (light blue in this figure) achieves little more robustness in $\overline{GT}_d(\delta)$ than that of either independence (Figure 3) or common marginals (orange line in this figure) on their own.

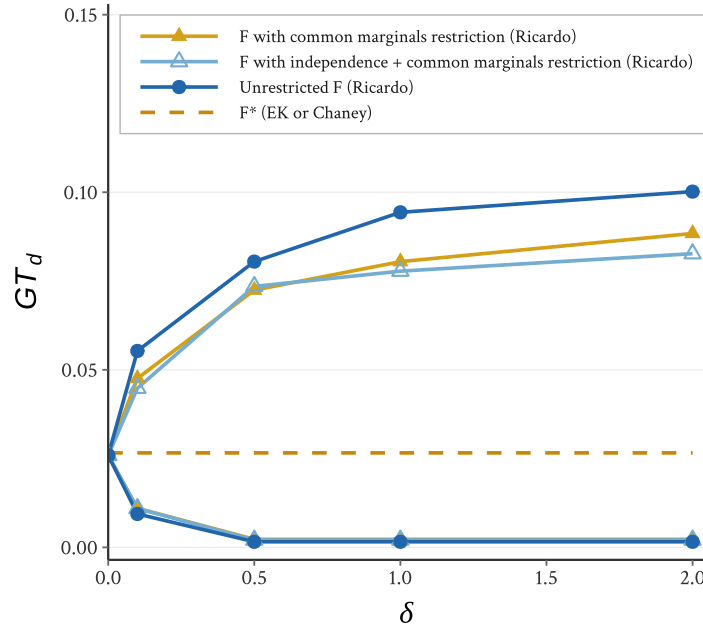


Figure 5: Ricardian gains from trade with independence and common marginals

5.6 Summary

Stepping back, the combined lessons from the restrictions imposed in Figures 3 through 5 imply that gains from trade in the Ricardian model can be many times larger or smaller than they are in the workhorse Eaton and Kortum (2002) case when one departs from that model's productivity distribution F^* in *either* of the two directions available. One direction is to relax independence, as we have seen in the case with Frechet marginals but unrestricted dependence in Figure 3. The other direction is to relax the Frechet marginals, as we have seen in the case with independence and common but otherwise unrestricted marginals in Figure 5. That is, at least as far as gains from trade are concerned, there are important ways to deviate from F^* , even in the small neighborhood of $\delta = 2$ and while matching all available data and a standard orthogonality condition, by perturbing either

the common marginal that countries share or the extent to which countries draw their productivities in dependent ways.

6 Concluding Remarks

Three modern revolutions have taken place in the study of international trade. A first is the desire to answer counterfactual questions using quantitative simulations. A second is the incorporation of micro-level production heterogeneity across goods and countries, either in Ricardian models a la [Eaton and Kortum \(2002\)](#) or monopolistically competitive models a la [Melitz \(2003\)](#). And a third is the desire to ensure that models exactly match all baseline trade flows, and their implied empirical patterns such as the fit of the gravity equation or the magnitude of the trade elasticity, before being used for counterfactual simulations away from the baseline scenario.

The goal of this paper has been to explore the robustness of this three-part revolution to variations in the assumed distribution of cross-country and cross-good heterogeneity in production technologies. Workhorse models such as [Eaton and Kortum \(2002\)](#) and [Chaney \(2008\)](#) assume that this heterogeneity is drawn from particular distributions that have been chosen largely for their tractability and alignment with gravity model traditions. But given the impossibility of direct evidence about these distributions—since what matters for key counterfactual questions such as the gains from trade is the distribution of a country’s productivity among goods it does not produce—it seems important to probe the sensitivity of model predictions to variations in the underlying distribution of assumed production heterogeneity.

Our results convey a strong sense in which quantitative trade models are highly sensitive to the underlying distribution of production heterogeneity that is assumed. And this is true despite ensuring that every distribution of heterogeneity that we consider is capable of generating, in full general equilibrium, the same values for all bilateral trade flows as occurs at the empirical baseline, as well as satisfying the same orthogonality conditions that are typically used for parameter estimation. Indeed, this finding continues to hold even when one maintains components of the workhorse distributions—e.g. that countries’ draws are independent but with arbitrary country-specific marginals, or independent with arbitrary but common marginals, or with benchmark common marginals but arbitrary amounts of (positive or negative) dependence.

A natural direction for future work is to extend the methods in this paper to a wider set of model features. In principle, our model can be augmented to include elements such as domestic input-output linkages, international supply chains, multiple sectors,

and multiple factors—all of which typically increase the gains from trade in quantitative models, implying that their omission here has been conservative. Likewise, phenomena such as factor mobility and consumer demand are typically modeled via distributional assumptions about the heterogeneity of households' preferences. The tools developed in this paper could be applied to such settings as well.

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A Theory Appendix

Throughout this Appendix we bound the gains from trade of a single country, which we label j . We write $\mathcal{A}(F)$ for the set of productivity shifters $\{A\}$ that rationalize the data under the distribution F (unique up to the normalization $P_d = 1$), and $\mathcal{V}$ for the distributions admitting a strictly positive density with a finite $(\sigma - 1)$ -th moment. We denote $\tau_{od} \equiv 1 + t_{od}$ and $V_{\tau\tau} \equiv \text{Var}(\Delta_o \Delta_d \ln \tau)$. For any bilateral variable M write

$$C_M \equiv \text{Cov}(\Delta_o \Delta_d M, \Delta_o \Delta_d \ln \tau),$$

A.1 Proof of Proposition 1(a)

Proposition 1(a) (Bounds for competitive model). *Suppose that the baseline trade data satisfy, for country j of interest, the condition:*

$$\lambda_{jj}/\lambda_{jd} \geq \lambda_{oj}/\lambda_{od} \quad \forall o, d \neq j \quad (\text{C1})$$

Consider all possible productivity distributions F (i.e. $\delta = \infty$). Then the smallest and largest possible values for j 's gains from trade in the competitive model, as defined in program (23), are given by

$$\underline{GT}_j(\delta = \infty) = 0 \quad \text{and} \quad \overline{GT}_j(\delta = \infty) = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (38)$$

Proof. Normalise $w_j = w'_j = 1$ and hold the deficit fixed as a share of income, so that by (12) and (2),

$$GT_j = 1 - \left(\frac{\gamma'_j}{\gamma_j} \right)^{1/(\sigma-1)}, \quad (39)$$

where γ_j and γ'_j are the factual and autarky values of the price aggregate (2) for j . Because GT_j is strictly decreasing in the autarky-to-trade ratio γ'_j/γ_j , it suffices to prove that in each model γ'_j/γ_j ranges over $(\lambda_{jj}, 1)$, with both endpoints sharp; the bounds on GT_j then follow from (39).

Setup. For a good ω and structural matrix A define the competitiveness of origin o in market d ,

$$\Phi_{od}(\omega) \equiv MC_{od}(\omega)^{1-\sigma} = \kappa_{od}^{1-\sigma} z_o(\omega)^{\sigma-1}, \quad \kappa_{od} \equiv \frac{w_o \tau_{od}}{A_{od}}.$$

By (4) the buyer in d purchases from $\arg \max_o \Phi_{od}$, so,

$$\gamma_d = \mathbb{E}_F \left[\max_o \Phi_{od} \right], \quad \lambda_{od} = \frac{\mathbb{E}_F [\Phi_{od} \mathbf{1}\{o \text{ wins } d\}]}{\gamma_d},$$

where “ o wins d ” is the event $\{\Phi_{od} \geq \Phi_{cd} \forall c\}$ (ties have F -measure zero and are ignored). In autarky j produces every good itself, so $\gamma'_j = \mathbb{E}_F[\Phi_{jj}]$.

Step 1 (universal bounds). Fix any $F \in \mathcal{V}$ with $A = \mathcal{A}(F)$. Pointwise, $\max_o \Phi_{oj} \geq \Phi_{jj} \geq 0$. The event $\{j \text{ loses } j\} = \{\exists o \neq j : \Phi_{oj} > \Phi_{jj}\}$ has strictly positive F -measure: otherwise $\mathbb{E}_F[\Phi_{oj} \mathbb{1}\{o \text{ wins } j\}] = 0$ for every $o \neq j$, forcing $\sum_{o \neq j} \lambda_{oj} = 0$, contradicting $\lambda_{jj} < 1$. On that event $\max_o \Phi_{oj} > \Phi_{jj}$ and $\Phi_{jj} > 0$ ($z_j > 0$ a.s. with positive density), whence

$$\mathbb{E}_F[\max_o \Phi_{oj}] > \mathbb{E}_F[\Phi_{jj}] \implies \gamma'_j / \gamma_j < 1. \quad (40)$$

Moreover $\lambda_{jj} \gamma_j = \mathbb{E}_F[\Phi_{jj} \mathbb{1}\{j \text{ wins } j\}] < \mathbb{E}_F[\Phi_{jj}] = \gamma'_j$, where the inequality is strict because

$\mathbb{E}_F[\Phi_{jj} \mathbb{1}\{j \text{ loses } j\}] > 0$ by the same argument; hence $\gamma'_j / \gamma_j > \lambda_{jj}$. Combining, $\gamma'_j / \gamma_j \in (\lambda_{jj}, 1)$, i.e.

$$0 < GT_j < 1 - \lambda_{jj}^{1/(\sigma-1)} \quad \text{for every } F \in \mathcal{V}. \quad (41)$$

Step 2 (thresholds and the losing boundary). Hold the productivity shifters fixed at the Frechet baseline $A^* = \mathcal{A}(F^*)$; by construction these satisfy the orthogonality restriction (19). Given competitors' draws $z_{-j} = (z_o)_{o \neq j}$, origin j wins market d iff $\Phi_{jd} \geq \Phi_{od}$ for all $o \neq j$, i.e. iff

$$z_j^{\sigma-1} \geq T_d(z_{-j}) \equiv \max_{o \neq j} \left(\frac{\kappa_{od}}{\kappa_{jd}} \right)^{1-\sigma} z_o^{\sigma-1},$$

for all κ evaluated at A^* . Define the losing boundary $B(z_{-j}) \equiv (\min_d T_d(z_{-j}))^{1/(\sigma-1)}$; that is, for $z_j < B$ origin j wins no market, and for $z_j \geq B$ it wins at least one.

Step 3 (Condition C1 pins the boundary at the home market). Under F^* , we have: $\lambda_{od} / \lambda_{jd} = (\kappa_{od} / \kappa_{jd})^{-\theta^*}$ for all o, d . Condition C1 is $\lambda_{jj} \lambda_{od} \geq \lambda_{oj} \lambda_{jd}$, or $\lambda_{od} / \lambda_{jd} \geq \lambda_{oj} / \lambda_{jj}$, which is equivalent to $(\kappa_{od} / \kappa_{jd})^{1-\sigma} \geq (\kappa_{oj} / \kappa_{jj})^{1-\sigma}$. Multiplying by $z_o^{\sigma-1} > 0$ and taking $\max_{o \neq j}$ we have,

$$T_j(z_{-j}) \leq T_d(z_{-j}) \quad \text{for all } d \neq j, z_{-j}, \quad \text{hence } B(z_{-j})^{\sigma-1} = T_j(z_{-j}). \quad (42)$$

In particular j wins its own market iff $z_j \geq B(z_{-j})$.

Step 4 (invariance of the data under reallocation below B). Let F have the same z_{-j} -marginal as F^* and a conditional law of z_j given z_{-j} that agrees with F^* on $[B(z_{-j}), \infty)$; the total conditional mass on $(0, B)$ is then automatically that of F^* . We claim every share is unchanged, so $\mathcal{A}(F) = A^*$. Condition on z_{-j} and abbreviate $B = B(z_{-j})$. Since $\max_{o \neq j} \Phi_{od} = \kappa_{jd}^{1-\sigma} T_d$, we have $\max_c \Phi_{cd} = \kappa_{jd}^{1-\sigma} \max\{z_j^{\sigma-1}, T_d\}$, and $T_d \geq T_j = B^{\sigma-1}$ by (42); thus

$$\mathbb{E}_F[\max_c \Phi_{cd} \mid z_{-j}] = \kappa_{jd}^{1-\sigma} \left(T_d \mathbb{P}(z_j < T_d^{1/(\sigma-1)} \mid z_{-j}) + \mathbb{E}[z_j^{\sigma-1} \mathbb{1}\{z_j \geq T_d^{1/(\sigma-1)}\} \mid z_{-j}] \right).$$

Because $T_d^{1/(\sigma-1)} \geq B$, both the probability $\mathbb{P}(z_j < T_d^{1/(\sigma-1)}) = \mathbb{P}(z_j < B) + \mathbb{P}(B \leq z_j < T_d^{1/(\sigma-1)})$ and the truncated expectation over $[T_d^{1/(\sigma-1)}, \infty) \subseteq [B, \infty)$ depend on the conditional law only through its total mass on $(0, B)$ and its restriction to $[B, \infty)$, both preserved; so $\gamma_d = \mathbb{E}_F[\max_c \Phi_{cd}]$ is unchanged. For the numerator with $o = j$, $\mathbb{E}_F[\Phi_{jd} \mathbb{1}\{j \text{ wins } d\} \mid z_{-j}] = \kappa_{jd}^{1-\sigma} \mathbb{E}[z_j^{\sigma-1} \mathbb{1}\{z_j \geq T_d^{1/(\sigma-1)}\} \mid z_{-j}]$ is unchanged for the same reason. For $o \neq j$, let $\mathcal{Z}_o(d) = \{z_{-j} : \Phi_{od} \geq \Phi_{cd} \forall c \neq j\}$, which is a function of z_{-j} only. On $\mathcal{Z}_o(d)$, $T_d = (\kappa_{od}/\kappa_{jd})^{1-\sigma} z_o^{\sigma-1}$ and o wins d iff additionally $z_j^{\sigma-1} \leq T_d$, so

$$\mathbb{E}_F[\Phi_{od} \mathbb{1}\{o \text{ wins } d\}] = \kappa_{od}^{1-\sigma} \mathbb{E}_F[z_o^{\sigma-1} \mathbb{1}_{\mathcal{Z}_o(d)} \mathbb{P}(z_j \leq T_d^{1/(\sigma-1)} \mid z_{-j})],$$

again unchanged since $T_d^{1/(\sigma-1)} \geq B$. Hence all λ_{od} are unchanged and $\mathcal{A}(F) = A^*$. This means that F matches all data and satisfies (19).

Step 5 (the $[0, B(z_{-j})]$ contribution to GT_j). By Step 3, j wins j iff $z_j \geq B$, so $\lambda_{jj}\gamma_j = \kappa_{jj}^{1-\sigma} \mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{z_j \geq B\}]$ and therefore

$$\gamma'_j = \mathbb{E}_F[\Phi_{jj}] = \lambda_{jj}\gamma_j + S(F), \quad S(F) \equiv \kappa_{jj}^{1-\sigma} \mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{z_j < B\}] \geq 0, \quad (43)$$

$$\frac{\gamma'_j}{\gamma_j} = \lambda_{jj} + \frac{S(F)}{\gamma_j}, \quad \frac{S(F)}{\gamma_j} \in [0, 1 - \lambda_{jj}]. \quad (44)$$

Decomposing $\gamma_j = \mathbb{E}_F[\max_o \Phi_{oj}] = \mathbb{E}_{F^*}[\max_o \Phi_{oj}]$ over $\{z_j \geq B\}$ (winner j , contributing $\lambda_{jj}\gamma_j$) and $\{z_j < B\}$ (where, by (42), $\max_o \Phi_{oj} = \kappa_{jj}^{1-\sigma} T_j = \kappa_{jj}^{1-\sigma} B^{\sigma-1}$) gives the companion identity

$$\kappa_{jj}^{1-\sigma} \mathbb{E}_{F^*}[B^{\sigma-1} \mathbb{1}\{z_j < B\}] = (1 - \lambda_{jj})\gamma_j. \quad (45)$$

Step 6 (upper bound is sharp). For $\alpha \in (0, 1)$ let F_α reallocate the below- B mass toward 0:

$$f_\alpha(z \mid z_{-j}) = \begin{cases} (1 - \alpha) p(z_{-j}) B^{-(1-\alpha)} z^{-\alpha}, & 0 < z < B, \\ f^*(z \mid z_{-j}), & z \geq B, \end{cases} \quad p(z_{-j}) \equiv \mathbb{P}^{F^*}(z_j < B \mid z_{-j}).$$

This is a density: $\int_0^B (1 - \alpha) p B^{-(1-\alpha)} z^{-\alpha} dz = p B^{-(1-\alpha)} B^{1-\alpha} = p$ (finite as $\alpha < 1$), plus mass $1 - p$ on $[B, \infty)$; it is strictly positive and agrees with f^* on $[B, \infty)$ with below- B mass p , so Step 4 gives $\mathcal{A}(F_\alpha) = A^*$. Its below- B $(\sigma - 1)$ -th moment is $\int_0^B z^{\sigma-1} f_\alpha dz = (1 - \alpha) p B^{-(1-\alpha)} \frac{B^{\sigma-\alpha}}{\sigma-\alpha} = \frac{1-\alpha}{\sigma-\alpha} p B^{\sigma-1}$ (finite as $\sigma - \alpha > 0$), so $F_\alpha \in \mathcal{V}$ and, by (45), $S(F_\alpha) =$

$\frac{1-\alpha}{\sigma-\alpha}(1-\lambda_{jj})\gamma_j$. Hence by (44)

$$\frac{\gamma'_j}{\gamma_j} = \lambda_{jj} + \frac{1-\alpha}{\sigma-\alpha}(1-\lambda_{jj}),$$

which, since $\frac{d}{d\alpha} \frac{1-\alpha}{\sigma-\alpha} = \frac{1-\sigma}{(\sigma-\alpha)^2} < 0$, decreases to λ_{jj} as $\alpha \uparrow 1$. By (39), $GT_j(F_\alpha) \uparrow 1 - \lambda_{jj}^{1/(\sigma-1)}$. With the strict ceiling (41), $\sup GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$, attained in the limit by F^+ (all below- B mass at 0), the case where j cannot reproduce its imports.

Step 7 (lower bound is sharp). For $n \in \mathbb{N}$ let F_n concentrate the below- B mass just under B : $f_n(z | z_{-j}) = n p(z_{-j}) B^{-n} z^{n-1}$ on $(0, B)$ and f^* on $[B, \infty)$; this is a density with below- B mass $\int_0^B n p B^{-n} z^{n-1} dz = p$, and below- B $(\sigma-1)$ -th moment $\int_0^B z^{\sigma-1} n p B^{-n} z^{n-1} dz = \frac{n}{n+\sigma-1} p B^{\sigma-1}$, so $F_n \in \mathcal{V}$ and $\mathcal{A}(F_n) = A^*$. Then $S(F_n) = \frac{n}{n+\sigma-1}(1-\lambda_{jj})\gamma_j$ and $\gamma'_j/\gamma_j = \lambda_{jj} + \frac{n}{n+\sigma-1}(1-\lambda_{jj}) \uparrow 1$ as $n \rightarrow \infty$, so $GT_j(F_n) \downarrow 0$. With (41), $\inf GT_j = 0$, attained in the limit by F^- (below- B mass at B^-), the case where j reproduces each imported good at the winning foreign price. Both F_α, F_n are absolutely continuous w.r.t. F^* ; this proves the bounds and their sharpness in the competitive model. $\square$

A.2 Proof of proposition 1(b)

Proposition 1(b) (Bounds for monopolistic competition model). *Suppose that the baseline tariff data satisfy the condition that,*

$$\exists i_1 \neq i_2, \quad \Delta_o \Delta_d \ln(1 + t_{i_1 i_2}) > 0 \quad (C2)$$

Consider all possible productivity distributions F (i.e. $\delta = \infty$). Then the smallest and largest possible values for j 's gains from trade in the monopolistic competition model, as defined in program (24), are given by

$$\underline{GT}_j(\delta = \infty) = 0 \quad \text{and} \quad \overline{GT}_j(\delta = \infty) = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (46)$$

Proof. Normalise $w_j = w'_j = 1$ and hold the deficit fixed as a share of income; then by (12)–(2) autarky expenditure equals factual expenditure, $E'_j = E_j$, and

$$GT_j = 1 - \left(\frac{\gamma'_j}{\gamma_j} \right)^{1/(\sigma-1)}. \quad (47)$$

It suffices to show that γ'_j/γ_j ranges over $(\lambda_{jj}, 1)$, with both endpoints sharp. Throughout, the counterfactual obeys the two restrictions of program (24): the mass of entrants is

unchanged, $N_o = N'_o$, and every home firm produces in autarky, $(\hat{z}_{jj})' = 1$.

Setup. Write $Y \equiv z^{\sigma-1}$ and, for the law F_o of Y at origin o ,

$$\mathcal{E}_F(k) \equiv \mathbb{E}_F[Y \mathbb{1}\{Y > k\}], \quad \bar{E}_F \equiv \mathbb{E}_F[Y] = \mathcal{E}_F(1).$$

The home marginal F_j is free in $\mathcal{V}$; the foreign marginals are i.i.d. Pareto on $(1, \infty)$ with shape $s \equiv \theta^*/(\sigma - 1) > 1$ (without loss for GT_j , which depends on the foreign blocks only through aggregates).

Step 1 (welfare under $N_o = N'_o$). A home firm with draw z charges $p_{jj} = \frac{\sigma}{\sigma-1} w_j / (A_{jj} z)$ (since there are no tariffs domestically), so $p_{jj}^{1-\sigma} = C Y$ with $C \equiv (\frac{\sigma}{\sigma-1} w_j / A_{jj})^{1-\sigma} > 0$, and it serves the home market iff $Y > k$, where $k \equiv \hat{z}_{jj}^{\sigma-1}$ is the factual home cutoff. By (15) the home contribution to the factual price aggregate is $N_j C \mathcal{E}_{F_j}(k)$, and being the share λ_{jj} of γ_j ,

$$\lambda_{jj} \gamma_j = N_j C \mathcal{E}_{F_j}(k). \quad (48)$$

In autarky j consumes only home varieties; with the entrant mass held at $N'_j = N_j$, all producing ($k' = 1$), and $w'_j = w_j$,

$$\gamma'_j = N_j C \mathcal{E}_{F_j}(1) = N_j C \bar{E}_{F_j}. \quad (49)$$

Dividing (49) by (48) (the common factor $N_j C$ cancels, as does E_j via $E'_j = E_j$),

$$\frac{\gamma'_j}{\gamma_j} = \frac{\lambda_{jj} \bar{E}_{F_j}}{\mathcal{E}_{F_j}(k)}. \quad (50)$$

Step 2 (universal bounds). Two inequalities pin down the range of (50). First, $\mathcal{E}_{F_j}(k) \leq \mathcal{E}_{F_j}(1) = \bar{E}_{F_j}$ for $k \geq 1$, so

$$\frac{\gamma'_j}{\gamma_j} \geq \lambda_{jj} \implies GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}. \quad (51)$$

Second, $(\hat{z}_{jj})' = 1$ requires the least-productive home firm ($Y = 1$) to exactly break even in autarky, $\frac{1}{\sigma} X'_{jj}(1) = w_j f_{jj}$. Using $X'_{jj}(Y) = E_j (\gamma'_j)^{-1} p_{jj}^{1-\sigma} = E_j C Y / \gamma'_j$ with (49), $X'_{jj}(1) = E_j / (N_j \bar{E}_{F_j})$; while the factual zero-cutoff-profit condition $\frac{1}{\sigma} X_{jj}(k) = w_j f_{jj}$ with (48) gives $f_{jj} = k \lambda_{jj} E_j / (\sigma N_j \mathcal{E}_{F_j}(k))$. The break-even equality is therefore $\mathcal{E}_{F_j}(k) = k \lambda_{jj} \bar{E}_{F_j}$, so by (50)

$$\frac{\gamma'_j}{\gamma_j} = \frac{1}{k} \leq 1 \implies GT_j \geq 0. \quad (52)$$

Step 3 (sharpness). Take the home marginal Pareto with shape $m > 1$: $\mathcal{E}_{F_j}(k) = \frac{m}{m-1}k^{1-m}$, $\bar{E}_{F_j} = \frac{m}{m-1}$, so $\bar{E}_{F_j}/\mathcal{E}_{F_j}(k) = k^{m-1}$ and (50) reads $\gamma'_j/\gamma_j = \lambda_{jj}k^{m-1}$, while the break-even bound (52) becomes $k \leq \lambda_{jj}^{-1/m}$.

Upper bound: choose the home fixed cost so the factual cutoff sits at the floor, $k = 1$ (all home firms already produce at the factual equilibrium); then $\gamma'_j/\gamma_j = \lambda_{jj}$ and $GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$, the attaining F^+ .

Lower bound: let $m \rightarrow \infty$ at the break-even cutoff $k = \lambda_{jj}^{-1/m}$, so the home marginal concentrates at the floor; then $\gamma'_j/\gamma_j = \lambda_{jj}^{1/m} \uparrow 1$ and $GT_j = 1 - \lambda_{jj}^{1/(m(\sigma-1))} \downarrow 0$, the attaining F^- .

Step 4 (feasibility of the gravity moments). It remains to exhibit, for each F_j above, a full array (A, f, f_E, N) that reproduces every bilateral flow and satisfies both orthogonality restrictions (19) and (20). Normalize $\gamma_d = 1$. A firm from o has sales $E_d p_{od}^{1-\sigma} = E_d Y/B_{od}$, $B_{od} \equiv (\frac{\sigma}{\sigma-1}w_o\tau_{od}/A_{od})^{\sigma-1}$ and operates iff $Y > k_{od} = B_{od}\sigma w_o f_{od}/E_d$; the trade share and fixed cost invert to

$$B_{od} = \frac{N_o \mathcal{E}_{F_o}(k_{od})}{\lambda_{od}}, \quad f_{od} = \frac{k_{od} \lambda_{od} E_d}{\sigma w_o N_o \mathcal{E}_{F_o}(k_{od})}. \quad (53)$$

These invert cell by cell: for any cutoff $k_{od} \geq 1$ the first relation in (53) fixes B_{od} and, equivalently, A_{od} . The second fixes f_{od} ; hence $(A_{od}, f_{od}) \mapsto (\lambda_{od}, k_{od})$ is a bijection at each cell, so the cutoff matrix $\{k_{od}\}$ is free. The origin scalars N_o (and f_{oE}) are fixed by the origins' free-entry conditions and enter only as origin-only terms. Writing $\mu_{od} \equiv \Delta_o \Delta_d \ln k_{od}$ and applying $\Delta_o \Delta_d$ to the logs of (53)

$$\Delta_o \Delta_d \ln A_{od} = \frac{1}{\sigma-1} \Delta_o \Delta_d \ln \lambda_{od} + \Delta_o \Delta_d \ln \tau_{od} + \frac{s-1}{\sigma-1} \mu_{od}, \quad (54)$$

$$\Delta_o \Delta_d \ln f_{od} = s \mu_{od} + \Delta_o \Delta_d \ln \lambda_{od}. \quad (55)$$

By the orthogonality moment (18), $C_\lambda = -\theta^* V_{\tau\tau} = -s(\sigma-1)V_{\tau\tau}$, so calculating the covariance of (54)–(55) and $\Delta_o \Delta_d \ln \tau$ yields (since $s > 1$) that both moments vanish iff the single equation

$$\text{Cov}(\mu, \Delta_o \Delta_d \ln \tau) = (\sigma-1)V_{\tau\tau} \quad (56)$$

holds. Set the home row $\ln k_{jd} = \ln k$ (constant in d) and the foreign diagonal cutoffs to the values pinned down by the foreign blocks' own free entry conditions (which are pure origin effects); place a single off-diagonal adjustment at the pivot (i_1, i_2) of Condition C2, $\nu_{i_1 i_2} \equiv (\sigma-1)V_{\tau\tau}/\Delta_o \Delta_d \ln \tau_{i_1 i_2} > 0$. All entries of μ other than the pivot are origin- or destination-only, hence simplified by $\Delta_o \Delta_d$, so $\text{Cov}(\mu, \Delta_o \Delta_d \ln \tau) = \nu_{i_1 i_2} \Delta_o \Delta_d \ln \tau_{i_1 i_2} =$

$(\sigma - 1)V_{\tau\tau}$, which is (56); every cutoff is interior. Thus for every $F_j \in \mathcal{V}$ the array (A, f, f_E, N) reproduces all bilateral flows, satisfies both orthogonality restrictions, and leaves all home firms producing in autarky, so it is feasible for program (24). With Steps 2–3 this proves $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$ and $\inf_F GT_j = 0$. $\square$

A.3 Proof of Proposition 2

Proposition 2 (Bounds for competitive model with independence). *Suppose that the base-line tariff and trade data satisfy the following condition: for Ξ a square matrix composed of elements Ξ_{od} that are zero apart from the j th row with elements given by $\frac{1}{1-\sigma} \ln \frac{\lambda_{jd}}{1-\lambda_{jd}}$, we have*

$$\mathbb{V}ar(\Delta_o \Delta_d \ln(1 + t_{od})) > \mathbb{C}ov(\Delta_o \Delta_d \Xi_{od}, \Delta_o \Delta_d \ln(1 + t_{od})) \quad (\text{C3})$$

Consider all possible productivity distributions F under which countries draw their productivities independently. Then the largest possible value for country j 's gains from trade in the competitive model, as defined in program (23), is given by

$$\overline{GT}_j^{\text{indep.}} = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (57)$$

Further, this value is approached arbitrarily closely by a sequence of distributions with Frechet marginals.

Proof. From the proof of Proposition 1(a), with the numeraire $w_j = w'_j = 1$,

$$GT_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}} (1 - L)^{-\frac{1}{\sigma-1}}, \quad L \equiv \frac{\mathbb{E}_F[z_j(\omega)^{\sigma-1} \mathbb{1}\{j \text{ loses } j\}]}{\mathbb{E}_F[z_j(\omega)^{\sigma-1}]}, \quad (\text{I1})$$

where “ j loses j ” is the event $\{z_j(\omega)^{\sigma-1} < \widetilde{M}\}$ with $\widetilde{M} \equiv \kappa_{jj}^{\sigma-1} \max_{o \neq j} \kappa_{oj}^{1-\sigma} z_o(\omega)^{\sigma-1}$ (in the notation of Appendix A.1, $\widetilde{M} = T_j$ and $L = S(F)/\gamma'_j$). Since $L \in [0, 1 - \lambda_{jj}]$, equation (12) gives the universal ceiling $GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}$, approached as $L \downarrow 0$, for every F . It remains to exhibit independent-Frechet, data-consistent models with $L \rightarrow 0$ and $C_{\ln A} = 0$.

Construction. For each $n \geq 1$ let every country draw independently, with j drawing from $\text{Frechet}(\alpha_n)$, $\alpha_n \equiv (\sigma - 1) + \frac{1}{n}$, and each competitor $o \neq j$ drawing from a common $\text{Frechet}(\gamma)$, $\gamma > \sigma - 1$ fixed below. Since $\alpha_n > \sigma - 1$, $\mathbb{E}[z_j^{\sigma-1}] < \infty$ (so $F_n \in \mathcal{V}$); the scales, equivalently the shifters $A^{(n)}$ through $\kappa_{od} = w_o \tau_{od} / A_{od}$, are pinned down in Step 3.

Steps 1 (the gravity moment by an intermediate-value argument). Let $\kappa^{(n)}(\alpha_n, \gamma)$ be the effective-cost matrix matching all shares at shapes (α_n, γ) (Step 2 shows it exists and is

unique), and put

$$G_n(\gamma) \equiv V_{\tau\tau} - C_{\ln \kappa^{(n)}(\alpha_n, \gamma)}. \quad (I2)$$

As $\ln \kappa_{od} = \ln w_o + \ln \tau_{od} - \ln A_{od}$ and $\Delta_o \Delta_d$ removes the origin-only $\ln w_o$, we have $\Delta_o \Delta_d \ln A_{od} = \Delta_o \Delta_d \ln \tau_{od} - \Delta_o \Delta_d \ln \kappa_{od}$, so $C_{\ln A} = V_{\tau\tau} - C_{\ln \kappa} = G_n$; thus the moment (19) is equivalent to $G_n(\gamma) = 0$.

Lower endpoint $G_n(\alpha_n) < 0$. At $\gamma = \alpha_n$ all shapes coincide, giving the single-elasticity **Eaton and Kortum (2002)** gravity model $\lambda_{od} = \kappa_{od}^{-\alpha_n} / \sum_{o'} \kappa_{o'd}^{-\alpha_n}$. Inverting, $\ln \kappa_{od} = -\frac{1}{\alpha_n} \ln \lambda_{od} + h_d$ with h_d destination-only, so $\Delta_o \Delta_d \ln \kappa_{od} = -\frac{1}{\alpha_n} \Delta_o \Delta_d \ln \lambda_{od}$ and $C_{\ln \kappa} = -\frac{1}{\alpha_n} C_{\ln \lambda} = \frac{\theta^*}{\alpha_n} V_{\tau\tau}$. Hence $G_n(\alpha_n) = V_{\tau\tau} (1 - \theta^* / \alpha_n) < 0$ for all n large enough that $\alpha_n < \theta^*$ (possible since $\theta^* > \sigma - 1$ and $\alpha_n \downarrow \sigma - 1$).

Upper endpoint $\lim_{\gamma \rightarrow \infty} G_n(\gamma) > 0$. For fixed γ the **Eaton and Kortum (2002)** closed form inverts the competitor entries to $\ln \kappa_{od}(\gamma) = \ln \Psi_d(\gamma) - \frac{1}{\gamma} \ln p_{od}$, $p_{od} \equiv \lambda_{od} / (1 - \lambda_{jd})$, $\Psi_d \equiv (\sum_{o \neq j} \kappa_{od}^{-\gamma})^{-1/\gamma}$, and the home entries to $\ln \kappa_{jd}(\gamma) = \ln \Psi_d(\gamma) - \frac{1}{\alpha_n} \ln s_d(\gamma)$, with $s_d(\gamma) \equiv (\Psi_d / \kappa_{jd})^{\alpha_n}$ pinned down by the home value share through a continuous, strictly increasing map $\lambda_{jd} = H_n(s_d; \gamma)$; explicitly, $H_n(s_d; \gamma) \equiv \mathbb{E}[Z \mathbb{1}\{Z > \tilde{M}\}] / \mathbb{E}[\max(Z, \tilde{M})]$ with $Z \sim \text{Frechet}(\theta_n, s_d^{1/\theta_n})$, $\tilde{M} \sim \text{Frechet}(\gamma / (\sigma - 1), 1)$, and $\theta_n \equiv \alpha_n / (\sigma - 1)$. The destination-only level $\ln \Psi_d$ is simplified by $\Delta_o \Delta_d$. As $\gamma \rightarrow \infty$ the competitor terms $-\frac{1}{\gamma} \ln p_{od} \rightarrow 0$ while the home row tends to $-\frac{1}{\alpha_n} \ln s_d^\infty$, so $\ln \kappa_{od}^{(n)} \rightarrow \ln \Psi_d + \Xi_{od}^{(n)}$ pointwise, with $\Xi_{od}^{(n)} \equiv -\frac{1}{\alpha_n} \ln s_d^\infty \mathbb{1}\{o = j\}$ and $s_d^\infty \equiv \lim_{\gamma \rightarrow \infty} s_d(\gamma) = (H_n^\infty)^{-1}(\lambda_{jd})$. As $\gamma \rightarrow \infty$ the competitors' maximum welfare-weighted productivity degenerates at its scale while j remains *Frechet*(α_n); $H_n(\cdot; \gamma)$ converges to a strictly increasing H_n^∞ , and the inverse of a continuous strictly monotone function depends continuously on it, so $s_d(\gamma) \rightarrow s_d^\infty$. Explicitly, $H_n^\infty(s_d) = s_d^{1/\theta_n} \gamma_{\text{low}} (1 - \frac{1}{\theta_n}, s_d) / (s_d^{1/\theta_n} \gamma_{\text{low}} (1 - \frac{1}{\theta_n}, s_d) + e^{-s_d})$, with $\gamma_{\text{low}}(a, x) \equiv \int_0^x v^{a-1} e^{-v} dv$ the lower incomplete gamma (which is finite since $\theta_n > 1$). Since $\Delta_o \Delta_d$ is a fixed linear (and hence continuous) operator simplifying the destination-only $\ln \Psi_d$,

$$\lim_{\gamma \rightarrow \infty} \Delta_o \Delta_d \ln \kappa_{od}^{(n)} = \Delta_o \Delta_d \Xi^{(n)}. \quad (58)$$

The leading-order expansion as $\alpha_n \downarrow \sigma - 1$ gives $\frac{1}{\alpha_n} \ln s_d^\infty = \frac{1}{\sigma - 1} \ln \frac{\lambda_{jd}}{1 - \lambda_{jd}}$ up to a destination-independent constant; on the home row that constant is a pure origin effect, simplified by $\Delta_o \Delta_d$, so $\Delta_o \Delta_d \Xi^{(n)} \rightarrow \Delta_o \Delta_d \Xi$ as $n \rightarrow \infty$. Therefore $\lim_{\gamma \rightarrow \infty} G_n(\gamma) = V_{\tau\tau} - C_{\Xi^{(n)}} \rightarrow V_{\tau\tau} - C_{\Xi} > 0$ by **C3**. Thus for all large n , G_n is continuous on $[\alpha_n, \infty)$ with $G_n(\alpha_n) < 0 < \lim_{\gamma \rightarrow \infty} G_n(\gamma)$, and the intermediate value theorem yields a finite root $\gamma_n^* \in (\alpha_n, \infty)$ with $G_n(\gamma_n^*) = 0$, i.e. the moment (19) is satisfied.

Step 2 (exact matching of all trade flows). Fix (α_n, γ_n^*) . Among competitors $c, k \neq j$ the

shares satisfy $\lambda_{cd}/\lambda_{kd} = (\kappa_{cd}/\kappa_{kd})^{-\gamma_n^*}$, while the home share λ_{jd} is smooth and strictly decreasing in κ_{jd} from 1 to 0. These supply $D(D-1)$ independent equations in the $D(D-1)$ unknown ratios of $\kappa^{(n)}$ (with the normalisation $\kappa_{jj} = 1$); by the inverse function theorem there is a unique $\kappa^{(n)}$, hence a unique $A_{od}^{(n)} = \kappa_{od}^{(n)}/(w_o\tau_{od})$, reproducing every bilateral flow. The moment (19) holds because $G_n(\gamma_n^*) = 0$, and the $(\sigma-1)$ -th moment is finite since $\alpha_n > \sigma-1$. Thus F_n is an admissible, independent, Fréchet-marginal model satisfying all constraints of program (23).

Step 4 (the welfare limit $L_n \rightarrow 0$). Let $Y \equiv z_j(\omega)^{\sigma-1} \sim \text{Frechet}(\theta_n)$, $\theta_n \equiv \alpha_n/(\sigma-1) = 1 + \frac{1}{n(\sigma-1)} \downarrow 1$, and let $\widetilde{M}_n$ be the competitor maximum in j 's market as in (I1); Y and $\widetilde{M}_n$ are independent. The two share identities at $d = j$,

$$\lambda_{jj} = \frac{\mathbb{E}[Y \mathbb{1}\{Y > \widetilde{M}_n\}]}{\mathbb{E}[\max(Y, \widetilde{M}_n)]}, \quad 1 - \lambda_{jj} = \frac{\mathbb{E}[\widetilde{M}_n \mathbb{1}\{\widetilde{M}_n \geq Y\}]}{\mathbb{E}[\max(Y, \widetilde{M}_n)]},$$

give $\mathbb{E}[\max(Y, \widetilde{M}_n)] = \mathbb{E}[Y](1 - L_n)/\lambda_{jj}$ and hence, using $\max(Y, \widetilde{M}_n) \geq \widetilde{M}_n$, $\mathbb{E}[\widetilde{M}_n]/\mathbb{E}[Y] \leq 1/\lambda_{jj}$. With $\widehat{Y}_n \equiv Y/\mathbb{E}[Y]$ ($\mathbb{E}[\widehat{Y}_n] = 1$) and $\widehat{M}_n \equiv \widetilde{M}_n/\mathbb{E}[Y]$ ($\mathbb{E}[\widehat{M}_n] \leq 1/\lambda_{jj}$), the loss fraction is $L_n = \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq \widehat{M}_n\}]$. By independence and Tonelli, for any fixed $K > 0$,

$$\begin{aligned} L_n &= \int_0^\infty y \mathbb{P}(\widehat{M}_n \geq y) f_{\widehat{Y}_n}(y) dy \leq \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] + \int_K^\infty y \cdot \frac{\mathbb{E}[\widehat{M}_n]}{y} f_{\widehat{Y}_n}(y) dy \\ &\leq \mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] + \frac{\mathbb{P}(\widehat{Y}_n > K)}{\lambda_{jj}}, \end{aligned}$$

applying Markov's inequality $\mathbb{P}(\widehat{M}_n \geq y) \leq \mathbb{E}[\widehat{M}_n]/y \leq 1/(\lambda_{jj}y)$ on $[K, \infty)$, where the factor y cancels against $1/y$. Fix K and let $n \rightarrow \infty$. Since $\mathbb{E}[Y] = \Gamma(1 - 1/\theta_n) \sim 1/(\theta_n - 1) = n(\sigma - 1) \rightarrow \infty$,

$$\mathbb{P}(\widehat{Y}_n > K) = 1 - \exp(-(K \mathbb{E}[Y])^{-\theta_n}) \rightarrow 0,$$

and the truncated mean $\mathbb{E}[\widehat{Y}_n \mathbb{1}\{\widehat{Y}_n \leq K\}] = \Gamma(1 - \frac{1}{\theta_n}, (K \mathbb{E}[Y])^{-\theta_n})/\Gamma(1 - \frac{1}{\theta_n}) \rightarrow 0$, because the bulk of the $\text{Frechet}(\theta_n)$ mass concentrates near 0 as $\theta_n \downarrow 1$ while its diverging mean is carried by a vanishing-probability upper tail. More precisely, the ratio is the regularized upper incomplete gamma $Q(s_n, x_n) = \Gamma(s_n, x_n)/\Gamma(s_n)$ with $s_n \equiv 1 - \frac{1}{\theta_n} \in (0, 1)$ and $x_n \equiv (K \mathbb{E}[Y])^{-\theta_n}$: as $n \rightarrow \infty$, $s_n \downarrow 0$ and $x_n \downarrow 0$; the expansion $\gamma_{\text{low}}(s, x) = \frac{x^s}{s}(1 + O(x))$ gives $Q(s, x) = 1 - \frac{x^s}{\Gamma(s+1)}(1 + O(x))$, and since $\theta_n s_n = \theta_n - 1$, $x_n^{s_n} = \exp(-(\theta_n - 1) \ln(K \mathbb{E}[Y]))$ with $\mathbb{E}[Y] \sim (\theta_n - 1)^{-1}$, so $(\theta_n - 1) \ln(K \mathbb{E}[Y]) \rightarrow 0$; thus $x_n^{s_n} \rightarrow 1$, $\Gamma(s_n + 1) \rightarrow 1$, and $Q \rightarrow 0$. Both terms vanish for every fixed K , so $L_n \rightarrow 0$. By (I1), $GT_j^{(n)} = 1 - \lambda_{jj}^{1/(\sigma-1)}(1 -$

$L_n)^{-1/(\sigma-1)} \rightarrow 1 - \lambda_{jj}^{1/(\sigma-1)}$; with the universal ceiling this gives $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$, approached but not attained. $\square$

A.4 Proof of Proposition 3

Condition 1. Fix the country of interest j . There is a partition $\mathcal{P} = \{S_1, \dots, S_K\}$ of the competitor set $O \setminus \{j\}$ such that

$$\sum_{k: C_{\psi S_k} < 0} \text{Cov} \left(\Delta_o \Delta_d \psi^{S_k}, \Delta_o \Delta_d \ln(t_{od} + 1) \right) < \text{Var}(\Delta_o \Delta_d \ln(t_{od} + 1)) (\sigma - 1 - \theta^*). \quad (C^{\mathcal{P}})$$

With

$$\psi_{od}^S \equiv \mathbb{1}\{o \in S\} \ln \frac{\lambda_{od}}{\Lambda_{S,d}}, \quad \Lambda_{S,d} \equiv \sum_{o' \in S} \lambda_{o'd}$$

($\psi^S \equiv 0$ when $|S| = 1$).

Proposition 3 (Bounds for competitive model with common marginal distributions). Suppose that the baseline tariff and trade data satisfy the data condition $C^{\mathcal{P}}$. Consider all possible productivity distributions F under which countries draw their productivities from a common marginal distribution but with potential dependence across countries. Then the largest possible value for country j 's gains from trade in the competitive model, as defined in program (23), is given by

$$\overline{GT}_j^{\text{CM}} = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}}. \quad (59)$$

Further, this bound is approached arbitrarily closely by a distribution with Frechet(1, θ) marginals.

Proof. Step 1 (the nested construction and the covariance identity). Partition $O \setminus \{j\} = S_1 \cup \dots \cup S_K$ and fix $\rho = (\rho_1, \dots, \rho_K) \in [0, 1]^K$. Let (X_o) follow the two-level nested-logistic (GEV) max-stable law

$$\mathbb{P}(X \leq x) = \exp \left(-x_j^{-\beta} - \sum_{k=1}^K \left(\sum_{o \in S_k} x_o^{-\beta/(1-\rho_k)} \right)^{1-\rho_k} \right),$$

under which j and the blocks $S_1, \dots, S_K$ are independent at the top level and each block S_k is a logistic nest of dependence ρ_k . Every marginal is $\text{Frechet}(1, \beta)$, with $\beta \equiv \theta/(\sigma - 1)$, so $z_o = X_o^{1/(\sigma-1)} \sim \text{Frechet}(1, \theta)$; at $\rho = 0$ this is the i.i.d. law. Denote it $F_{\theta, \rho}^{\mathcal{P}}$. Writing $b_{od} \equiv \kappa_{od}^{1-\sigma}$, each offer is $\Phi_{od} = b_{od} X_o$. Substituting $x_o = \phi/b_{od}$ into the displayed CDF

(the event $\{\Phi_{od} \leq \phi \forall o\}$) collapses every argument to ϕ :

$$\mathbb{P}(\Phi_d^{\max} \leq \phi) = \exp(-\Theta_d \phi^{-\beta}), \quad \Theta_d = b_{jd}^\beta + \sum_k T_{S_k}(d), \quad T_{S_k}(d) = D_{kd}^{1-\rho_k}, \quad D_{kd} \equiv \sum_{o \in S_k} b_{od}^{\beta/(1-\rho_k)},$$

where the block term used $(\sum_{o \in S_k} (\phi/b_{od})^{-\beta/(1-\rho_k)})^{1-\rho_k} = \phi^{-\beta} D_{kd}^{1-\rho_k}$. Thus the global best offer is $\Phi_d^{\max} \sim \text{Frechet}(\Theta_d, \beta)$, the home offer is $\text{Frechet}(b_{jd}^\beta, \beta)$, and the block maximum $\max_{o \in S_k} \Phi_{od} \sim \text{Frechet}(T_{S_k}(d), \beta)$, all independent. For a GEV max-stable law the identity of the argmax is independent of the value of the maximum, so the expenditure share equals the choice probability; the nested-logit factorisation (nest choice $\times$ within-nest choice) gives

$$\lambda_{jd} = \frac{b_{jd}^\beta}{\Theta_d}, \quad \lambda_{od} = \underbrace{\frac{T_{S_k}(d)}{\Theta_d}}_{=\Lambda_{S_k,d}} \underbrace{\frac{b_{od}^{\beta/(1-\rho_k)}}{D_{kd}}}_{=\lambda_{od}/\Lambda_{S_k,d}} \quad (o \in S_k), \quad \Lambda_{S_k,d} \equiv \sum_{o \in S_k} \lambda_{od}.$$

Taking logs of the two competitor factors and using $b_{od}^\beta = \kappa_{od}^{(1-\sigma)\beta} = \kappa_{od}^{-\theta}$ (so $\frac{\beta}{1-\rho_k} \ln b_{od} = -\frac{\theta}{1-\rho_k} \ln \kappa_{od}$),

$$\psi_{od}^{S_k} = \ln \frac{\lambda_{od}}{\Lambda_{S_k,d}} = -\frac{\theta}{1-\rho_k} \ln \kappa_{od} - \ln D_{kd}, \quad \ln \Lambda_{S_k,d} = (1-\rho_k) \ln D_{kd} - \ln \Theta_d.$$

Since $\ln \lambda_{od} = \ln \Lambda_{S_k,d} + \psi_{od}^{S_k}$, the combination that eliminates $\ln D_{kd}$ is

$$\ln \lambda_{od} - \rho_k \psi_{od}^{S_k} = \ln \Lambda_{S_k,d} + (1-\rho_k) \psi_{od}^{S_k} = -\theta \ln \kappa_{od} - \ln \Theta_d.$$

The home relation $\lambda_{jd} = \kappa_{jd}^{-\theta}/\Theta_d$ inverts the shares to

$$\ln \kappa_{od} = -\frac{1}{\theta} (\ln \lambda_{od} - \rho_k \psi_{od}^{S_k}) - \frac{1}{\theta} \ln \Theta_d \quad (o \in S_k), \quad \ln \kappa_{jd} = -\frac{1}{\theta} \ln \lambda_{jd} - \frac{1}{\theta} \ln \Theta_d.$$

Since $\ln A_{od} = -\ln \kappa_{od} + \ln w_o + \ln \tau_{od}$ we have,

$$\Delta_o \Delta_d \ln A_{od} = \frac{1}{\theta} \Delta_o \Delta_d (\ln \lambda_{od} - \Psi_{od}^{\mathcal{P}, \rho}) + \Delta_o \Delta_d \ln \tau_{od}, \quad \Psi^{\mathcal{P}, \rho} \equiv \sum_k \rho_k \psi^{S_k}.$$

Pairing with $\Delta_o \Delta_d \ln \tau$ and using $C_{\ln \lambda} = -\theta^* V_{\tau\tau}$ gives the covariance identity, affine in ρ ,

$$K(\mathcal{A}(F_{\theta, \rho}^{\mathcal{P}})) = \frac{1}{\theta} \left(C_{\ln \lambda} - \sum_k \rho_k C_{\psi^{S_k}} \right) + V_{\tau\tau}.$$

Step 2 (clearing the manifold down to $\sigma - 1$). $K(\mathcal{A}(F_{\theta,\rho}^{\mathcal{P}})) = 0$ is the single linear equation

$$\sum_k \rho_k C_{\psi^{s_k}} = V_{\tau\tau} (\theta - \theta^*). \quad (60)$$

Set $\rho_k = 0$ on every block with $C_{\psi^{s_k}} \geq 0$ and let $\rho_k \in [0, 1)$ on the negative blocks; the left side then sweeps the interval $(\sum_{k: C_{\psi^{s_k}} < 0} C_{\psi^{s_k}}, 0]$, where (approached as those $\rho_k \uparrow 1$). For $\theta \in (\sigma - 1, \theta^*]$ the right side lies in $[V_{\tau\tau}(\sigma - 1 - \theta^*), 0]$; under condition $\mathcal{C}^{\mathcal{P}}$, $\sum_{k: C_{\psi^{s_k}} < 0} C_{\psi^{s_k}} < V_{\tau\tau}(\sigma - 1 - \theta^*)$, so the whole target interval is reached and (60) has a solution $\rho(\theta) \in [0, 1)^K$ for every $\theta \in (\sigma - 1, \theta^*]$ (e.g. uniform $\rho_k = V_{\tau\tau}(\theta - \theta^*) / \sum_{k: C_{\psi^{s_k}} < 0} C_{\psi^{s_k}}$ on the negative blocks). Each $F_{\theta,\rho(\theta)}^{\mathcal{P}}$ has common *Frechet*(1, θ) marginals, finite $(\sigma - 1)$ -th moments (since $\theta > \sigma - 1$), matches all bilateral trade flows by the inversion, and satisfies $K = 0$; it therefore obeys all constraints of program (23) with arbitrary (GEV) cross-country dependence.

Step 3 (welfare is the gravity value at θ , free of dependence). From the proof of Proposition 1(a),

$$GT_j = 1 - \lambda_{jj}^{\frac{1}{\sigma-1}} \left(\frac{\mathbb{E}_F[z_j(\omega)^{\sigma-1}]}{\mathbb{E}_F[z_j(\omega)^{\sigma-1} \mathbb{1}\{j \text{ wins } j\}]} \right)^{\frac{1}{\sigma-1}}.$$

In every distribution constructed above, z_j is drawn independently of $(z_o)_{o \neq j}$ and all marginals are *Frechet*(1, θ). The home offer $b_{jj}X_j \sim \text{Frechet}(b_{jj}^\beta, \beta)$ and the competitor maximum $\max_{o \neq j} b_{oj}X_o$ are independent, the latter *Frechet*(T, β) for some scale T . By the Eaton and Kortum (2002) value identity for two independent common-shape Frechet offers,

$$\mathbb{E}_F[z_j^{\sigma-1} \mathbb{1}\{j \text{ wins } j\}] / \mathbb{E}_F[z_j^{\sigma-1}] = \lambda_{jj}^{1-1/\beta} = \lambda_{jj}^{1-(\sigma-1)/\theta}, \text{ whence}$$

$$GT_j(F_{\theta,\rho(\theta)}^{\mathcal{P}}) = 1 - \lambda_{jj}^{1/\theta}.$$

GT_j is free of the dependence, which enters only through the scale T .

Step 4 (the limit). $GT_j(F_{\theta,\rho(\theta)}^{\mathcal{P}}) = 1 - \lambda_{jj}^{1/\theta} \uparrow 1 - \lambda_{jj}^{1/(\sigma-1)}$ as $\theta \downarrow \sigma - 1$. Combined with the universal ceiling $GT_j \leq 1 - \lambda_{jj}^{1/(\sigma-1)}$ (valid for every F), this gives $\sup_F GT_j = 1 - \lambda_{jj}^{1/(\sigma-1)}$. $\square$